

# CSCM Tool

## Controller Status Code Management Tool

# Test report

Results of the automated verification

| | |
|---|---|
| Version | 4.18.1 |
| Date | 2026-10-29 |
| Commit | a741076 |
| Source digest | bdd14c56645c7f8c0ce6df2b (SHA-256 of the source files) |

Generated from the Markdown documents of the repository by scripts/build_docs_pdf.py; do not edit by hand. Status symbols are written as words (OK, FAIL, yes), arrows and comparison signs as ASCII (->, <=, >=). The same version number is shown in the menu of the application and in /api/v1/health.

# 24. Test Report — CSCM Tool

> **Generated file — do not edit by hand.** Regenerate with `python cscm_tool/scripts/generate_test_report.py` after every software change (Definition of Done, 15-development-standards.md). It runs the complete pytest suite, measures coverage, runs the static quality gates, and appends the run to the history below.

## 1. Run summary

| | | |
|---|---|---|
| Verdict | PASS | |
| Generated (UTC) | 2026-10-29 01:18:38 | |
| Branch / commit | main @ a741076 +uncommitted changes | |
| Test order | file | |
| Tests | 1283 run — 1280 passed, 0 failed, 3 skipped | |
| Duration | 1749.2 s | |
| Line coverage (app/) | 93% (13857 of 14923 statements) | |
| Python / platform | 3.12.10 / Windows 11 | |
| pytest summary line | `functional: 1103 passed, 3 skipped, 9 deselected, 9 warnings in 1697.65s (0:28:17) | performance: 9 passed, 1 warning in 67.94s (0:01:07)` |

## 2. Results by suite

| Suite | Tests | Passed | Failed | Skipped | Time |
|---|---|---|---|---|---|
| unit | 74 | 74 | 0 | 0 | 5.2 s |
| integration | 761 | 761 | 0 | 0 | 1197.9 s |
| system | 439 | 436 | 0 | 3 | 478.6 s |
| performance | 9 | 9 | 0 | 0 | 67.5 s |

## 3. Static quality and security gates

| Gate | Result | Detail |
|---|---|---|
| black --check | PASS | 278 files would be left unchanged. |
| ruff check | PASS | All checks passed! |
| mypy | PASS | Success: no issues found in 163 source files |
| bandit -r app | PASS | [tester] WARNING nosec encountered (B324), but no failed test on line 76 |
| pip-audit | PASS | runtime packages with advisories: none; dev-only advisories (accepted): black, pytest, starlette |

## 4. Measured performance (NFR-001, NFR-004, TC-024, TC-025)

| Test | Measured |
|---|---|
| test_search_and_read_latency_at_20000_revisions | read_p95_ms = 9.2, revisions = 20000, search_median_ms = 86.5, search_p95_ms = 110.1, speed_factor = 1.0 |
| test_export_of_5000_records_within_target[csv-5] | export_seconds = 1.02, limit_seconds = 5, records = 5000, speed_factor = 1.0 |
| test_export_of_5000_records_within_target[json-5] | export_seconds = 1.3, limit_seconds = 5, records = 5000, speed_factor = 1.0 |
| test_export_of_5000_records_within_target[xlsx-15] | export_seconds = 3.21, limit_seconds = 15, records = 5000, speed_factor = 1.0 |
| test_export_of_5000_records_within_target[pdf-15] | export_seconds = 9.16, limit_seconds = 15, records = 5000, speed_factor = 1.0 |
| test_concurrent_identifier_allocation_in_one_group | allocation_p95_ms = 131.2, speed_factor = 1.0, submissions = 64 |
| test_fifty_concurrent_users_are_served_without_errors | p95_ms_50_users = 234.4, speed_factor = 1.0 |
| test_export_issues_a_constant_number_of_queries | statements_for_two_500_row_exports = 27 |

Targets: search p95 <= 500 ms and single read p95 <= 200 ms at 20,000 revisions; CSV/JSON export of 5,000 records <= 5 s, XLSX/PDF <= 15 s; 50 concurrent users keep the 500 ms search target; identifier allocation under contention correct and gap-free.

# 5. Coverage by module

| Module | Statements | Missed | Coverage |
|---|---|---|---|
| app/__init__.py | 84 | 0 | 100% |
| app/api/__init__.py | 0 | 0 | 100% |
| app/api/v1/__init__.py | 3 | 0 | 100% |
| app/api/v1/_errors.py | 20 | 2 | 90% |
| app/api/v1/audit_routes.py | 12 | 0 | 100% |
| app/api/v1/auth_routes.py | 60 | 4 | 93% |
| app/api/v1/bulk_routes.py | 92 | 0 | 100% |
| app/api/v1/change_request_routes.py | 136 | 9 | 93% |
| app/api/v1/domain_signoff_routes.py | 60 | 3 | 95% |
| app/api/v1/export_routes.py | 77 | 1 | 99% |
| app/api/v1/governance_routes.py | 244 | 3 | 99% |
| app/api/v1/health_routes.py | 30 | 2 | 93% |
| app/api/v1/import_routes.py | 93 | 11 | 88% |
| app/api/v1/lookup_routes.py | 70 | 1 | 99% |
| app/api/v1/mapping_routes.py | 131 | 7 | 95% |
| app/api/v1/operations_routes.py | 165 | 0 | 100% |
| app/api/v1/package_routes.py | 71 | 8 | 89% |
| app/api/v1/publication_routes.py | 196 | 2 | 99% |
| app/api/v1/release_routes.py | 79 | 8 | 90% |
| app/api/v1/revision_routes.py | 26 | 3 | 88% |
| app/api/v1/serializers.py | 58 | 0 | 100% |
| app/api/v1/st_import_routes.py | 72 | 2 | 97% |
| app/api/v1/status_code_routes.py | 212 | 18 | 92% |
| app/api/v1/user_routes.py | 96 | 6 | 94% |
| app/audit/__init__.py | 0 | 0 | 100% |
| app/audit_chain.py | 134 | 7 | 95% |
| app/audit_hash.py | 10 | 0 | 100% |
| app/auth/__init__.py | 0 | 0 | 100% |
| app/auth/decorators.py | 122 | 3 | 98% |
| app/auth/password.py | 30 | 0 | 100% |
| app/auth/providers.py | 32 | 1 | 97% |
| app/auth/rate_limit.py | 34 | 0 | 100% |
| app/auth/roles.py | 27 | 0 | 100% |
| app/auth/windows.py | 43 | 2 | 95% |
| app/branding.py | 6 | 0 | 100% |
| app/cli.py | 125 | 13 | 90% |
| app/config.py | 59 | 0 | 100% |
| app/db_modes.py | 15 | 0 | 100% |
| app/domain/__init__.py | 0 | 0 | 100% |
| app/domain/authz.py | 5 | 0 | 100% |
| app/domain/errors.py | 29 | 0 | 100% |
| app/domain/lifecycle.py | 19 | 0 | 100% |
| app/errors.py | 53 | 3 | 94% |
| app/export/__init__.py | 0 | 0 | 100% |
| app/export/csv_exporter.py | 17 | 0 | 100% |
| app/export/full_database_exporter.py | 22 | 1 | 95% |
| app/export/json_exporter.py | 3 | 0 | 100% |
| app/export/offline_bundle.py | 13 | 0 | 100% |
| app/export/pdf_exporter.py | 214 | 0 | 100% |

| Module | Statements | Missed | Coverage |
|---|---|---|---|
| app/export/xlsx_exporter.py | 22 | 0 | 100% |
| app/extensions.py | 9 | 0 | 100% |
| app/help_content.py | 6 | 0 | 100% |
| app/i18n.py | 100 | 9 | 91% |
| app/i18n_ar.py | 1 | 0 | 100% |
| app/i18n_de.py | 3 | 0 | 100% |
| app/i18n_eu.py | 4 | 0 | 100% |
| app/i18n_extra.py | 10 | 0 | 100% |
| app/i18n_generated.py | 13 | 0 | 100% |
| app/i18n_in.py | 4 | 0 | 100% |
| app/i18n_messages.py | 26 | 0 | 100% |
| app/importers/__init__.py | 0 | 0 | 100% |
| app/importers/glossary.py | 28 | 0 | 100% |
| app/importers/stcd.py | 227 | 5 | 98% |
| app/logging_config.py | 45 | 0 | 100% |
| app/models/__init__.py | 12 | 0 | 100% |
| app/models/audit.py | 16 | 0 | 100% |
| app/models/change_request.py | 136 | 0 | 100% |
| app/models/export_job.py | 38 | 0 | 100% |
| app/models/import_job.py | 30 | 0 | 100% |
| app/models/lookup.py | 87 | 0 | 100% |
| app/models/mapping.py | 69 | 0 | 100% |
| app/models/operations.py | 121 | 0 | 100% |
| app/models/release.py | 30 | 0 | 100% |
| app/models/status_code.py | 145 | 0 | 100% |
| app/models/terminology.py | 25 | 0 | 100% |
| app/models/translation.py | 42 | 0 | 100% |
| app/models/user.py | 33 | 0 | 100% |
| app/observability.py | 102 | 12 | 88% |
| app/repositories/__init__.py | 0 | 0 | 100% |
| app/repositories/audit_repository.py | 22 | 0 | 100% |
| app/repositories/base.py | 20 | 1 | 95% |
| app/repositories/change_request_repository.py | 26 | 0 | 100% |
| app/repositories/fts.py | 35 | 2 | 94% |
| app/repositories/release_repository.py | 4 | 0 | 100% |
| app/repositories/status_code_repository.py | 87 | 2 | 98% |
| app/repositories/user_repository.py | 33 | 1 | 97% |
| app/schema_sql.py | 45 | 1 | 98% |
| app/security_headers.py | 15 | 0 | 100% |
| app/services/__init__.py | 0 | 0 | 100% |
| app/services/attachment_service.py | 106 | 11 | 90% |
| app/services/audience_service.py | 31 | 1 | 97% |
| app/services/audit_service.py | 23 | 0 | 100% |
| app/services/auth_service.py | 108 | 9 | 92% |
| app/services/backup_service.py | 131 | 7 | 95% |
| app/services/baseline_service.py | 68 | 0 | 100% |
| app/services/bulk_service.py | 354 | 36 | 90% |
| app/services/change_request_service.py | 264 | 13 | 95% |
| app/services/change_set_service.py | 135 | 15 | 89% |
| app/services/connection_service.py | 131 | 5 | 96% |
| app/services/corpus_service.py | 75 | 3 | 96% |

| Module | Statements | Missed | Coverage |
|---|---|---|---|
| app/services/delegation_service.py | 86 | 6 | 93% |
| app/services/domain_signoff_service.py | 91 | 5 | 95% |
| app/services/duplicate_service.py | 41 | 1 | 98% |
| app/services/export_profile_service.py | 97 | 19 | 80% |
| app/services/export_service.py | 269 | 8 | 97% |
| app/services/field_review_service.py | 126 | 12 | 90% |
| app/services/health_report_service.py | 81 | 4 | 95% |
| app/services/hub_consolidation_service.py | 106 | 8 | 92% |
| app/services/import_service.py | 483 | 43 | 91% |
| app/services/integration_service.py | 192 | 7 | 96% |
| app/services/kpi_service.py | 72 | 7 | 90% |
| app/services/legacy_mapping_service.py | 73 | 6 | 92% |
| app/services/lookup_admin_service.py | 96 | 5 | 95% |
| app/services/lookup_service.py | 33 | 4 | 88% |
| app/services/mail_service.py | 30 | 3 | 90% |
| app/services/manifest_service.py | 46 | 1 | 98% |
| app/services/mapping_service.py | 636 | 49 | 92% |
| app/services/notification_service.py | 98 | 0 | 100% |
| app/services/numbering_service.py | 78 | 3 | 96% |
| app/services/occurrence_service.py | 88 | 7 | 92% |
| app/services/package_service.py | 199 | 27 | 86% |
| app/services/reference_service.py | 72 | 11 | 85% |
| app/services/release_notes_service.py | 115 | 9 | 92% |
| app/services/release_service.py | 136 | 6 | 96% |
| app/services/revision_diff_service.py | 35 | 0 | 100% |
| app/services/saved_search_service.py | 61 | 0 | 100% |
| app/services/settings_service.py | 74 | 6 | 92% |
| app/services/sla_service.py | 57 | 0 | 100% |
| app/services/software_version_service.py | 48 | 1 | 98% |
| app/services/st_import_service.py | 642 | 80 | 88% |
| app/services/status_code_service.py | 456 | 19 | 96% |
| app/services/terminology_service.py | 293 | 21 | 93% |
| app/services/translation_impact_service.py | 127 | 8 | 94% |
| app/services/translation_review_service.py | 362 | 33 | 91% |
| app/services/translation_service.py | 123 | 10 | 92% |
| app/services/user_service.py | 162 | 3 | 98% |
| app/services/validation.py | 88 | 11 | 88% |
| app/status_code_fields.py | 164 | 6 | 96% |
| app/terminology_seed.py | 14 | 1 | 93% |
| app/terminology_seed_process.py | 2 | 0 | 100% |
| app/terminology_seed_product.py | 1 | 0 | 100% |
| app/translation_catalogue.py | 141 | 0 | 100% |
| app/translation_precheck.py | 146 | 21 | 86% |
| app/ui/__init__.py | 73 | 8 | 89% |
| app/ui/admin_views.py | 170 | 21 | 88% |
| app/ui/auth_views.py | 95 | 26 | 73% |
| app/ui/bulk_views.py | 114 | 17 | 85% |
| app/ui/change_request_views.py | 192 | 22 | 89% |
| app/ui/dashboard_views.py | 34 | 0 | 100% |
| app/ui/export_views.py | 23 | 1 | 96% |
| app/ui/governance_views.py | 255 | 40 | 84% |

| Module | Statements | Missed | Coverage |
|---|---|---|---|
| app/ui/import_views.py | 134 | 14 | 90% |
| app/ui/insight_views.py | 118 | 17 | 86% |
| app/ui/mapping_views.py | 202 | 31 | 85% |
| app/ui/nav.py | 72 | 2 | 97% |
| app/ui/publication_views.py | 198 | 50 | 75% |
| app/ui/release_views.py | 95 | 29 | 69% |
| app/ui/status_code_views.py | 332 | 55 | 83% |
| app/ui/terminology_views.py | 114 | 5 | 96% |
| app/ui/translation_views.py | 138 | 4 | 97% |
| app/utils.py | 27 | 0 | 100% |
| app/version.py | 1 | 0 | 100% |

## 6. Traceability evidence check

`docs/20-traceability-matrix.md` cites 626 test names / test files as evidence. **626 resolve** to a test that exists; **0 are stale**.

## 6a. Requirement coverage

The specification defines **254** requirement ids (FR, NFR, BR in `03-srs.md`, SEC in `11-security-architecture.md`, AUD in `12-audit-compliance.md`). **216** are cited by at least one test (module docstring, test docstring or test name); **37** more are marked covered, with the tests named, in `20-traceability-matrix.md`; **1** are marked there as out of scope or needing a person (`NFR-002`). **0 are not accounted for**.

## 7. Failures

None.

## 8. Complete test inventory

Every executed test, grouped by file, with its outcome, duration, and the requirement IDs it cites (module/test docstring or name). Parametrised cases appear individually.

### tests/unit/test_lifecycle.py — 52 tests, 52 passed

TC-001 / TC-002 / BR-004: the 7-state lifecycle and its guard.

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_every_defined_transition_is_accepted[APPROVED-RELEASED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_defined_transition_is_accepted[DEPRECATED-ARCHIVED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_defined_transition_is_accepted[DEPRECATED-RELEASED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_defined_transition_is_accepted[DRAFT-REVIEW] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_defined_transition_is_accepted[PENDING_APPROVAL-APPROVED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_defined_transition_is_accepted[PENDING_APPROVAL-DEPRECATED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_defined_transition_is_accepted[PENDING_APPROVAL-DRAFT] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_defined_transition_is_accepted[REVIEW-DRAFT] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_defined_transition_is_accepted[REVIEW-PENDING_APPROVAL] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[APPROVED-APPROVED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[APPROVED-ARCHIVED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[APPROVED-DEPRECATED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[APPROVED-DRAFT] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[APPROVED-PENDING_APPROVAL] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[APPROVED-REVIEW] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[ARCHIVED-APPROVED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[ARCHIVED-ARCHIVED] | | BR-004, TC-001, TC-002 | 0.00 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_every_other_pairwise_transition_is_rejected[ARCHIVED-DEPRECATED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[ARCHIVED-DRAFT] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[ARCHIVED-PENDING_APPROVAL] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[ARCHIVED-RELEASED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[ARCHIVED-REVIEW] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[DEPRECATED-APPROVED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[DEPRECATED-DEPRECATED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[DEPRECATED-DRAFT] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[DEPRECATED-PENDING_APPROVAL] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[DEPRECATED-REVIEW] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[DRAFT-APPROVED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[DRAFT-ARCHIVED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[DRAFT-DEPRECATED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[DRAFT-DRAFT] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[DRAFT-PENDING_APPROVAL] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[DRAFT-RELEASED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[PENDING_APPROVAL-ARCHIVED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[PENDING_APPROVAL-PENDING_APPROVAL] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[PENDING_APPROVAL-RELEASED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[PENDING_APPROVAL-REVIEW] | | BR-004, TC-001, TC-002 | 0.01 s |
| OK | test_every_other_pairwise_transition_is_rejected[RELEASED-APPROVED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[RELEASED-ARCHIVED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[RELEASED-DEPRECATED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[RELEASED-DRAFT] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[RELEASED-PENDING_APPROVAL] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[RELEASED-RELEASED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[RELEASED-REVIEW] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[REVIEW-APPROVED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[REVIEW-ARCHIVED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[REVIEW-DEPRECATED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[REVIEW-RELEASED] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_every_other_pairwise_transition_is_rejected[REVIEW-REVIEW] | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_orm_guard_blocks_an_illegal_status_assignment | | BR-004, TC-001, TC-002 | 1.11 s |
| OK | test_terminal_and_gated_states_cannot_be_skipped | | BR-004, TC-001, TC-002 | 0.00 s |
| OK | test_there_are_exactly_seven_states | | BR-004, TC-001, TC-002 | 0.01 s |

## tests/unit/test_password.py — 5 tests, 5 passed

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_hash_and_verify_round_trip | | | 0.11 s |
| OK | test_policy_accepts_reasonable_password | | | 0.00 s |
| OK | test_policy_rejects_blocklisted_password | | | 0.00 s |
| OK | test_policy_rejects_short_password | | | 0.00 s |
| OK | test_verify_rejects_wrong_password | | | 0.12 s |

## tests/unit/test_rate_limit.py — 6 tests, 6 passed

SEC-004 / I-20: the login rate limiter, backed by the database.

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_refused_attempt_is_not_recorded | | NFR-009, SEC-004, TC-050 | 0.64 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_allows_attempts_under_the_limit | | NFR-009, SEC-004, TC-050 | 0.63 s |
| OK | test_attempts_age_out_of_the_window | | NFR-009, SEC-004, TC-050 | 0.63 s |
| OK | test_blocks_after_five_username_attempts | | NFR-009, SEC-004, TC-050 | 0.58 s |
| OK | test_blocks_after_ten_ip_attempts_across_usernames | | NFR-009, SEC-004, TC-050 | 0.57 s |
| OK | test_username_limit_is_case_insensitive | | NFR-009, SEC-004, TC-050 | 0.61 s |

## tests/unit/test_validation.py — 11 tests, 11 passed

FR-022, FR-027 field rules from docs/06-data-dictionary.md, v3

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_description_too_short | | FR-022, FR-027 | 0.00 s |
| OK | test_number_of_times_repeated_must_be_at_least_one | | FR-022, FR-027 | 0.01 s |
| OK | test_repeated_over_the_course_format | | FR-022, FR-027 | 0.01 s |
| OK | test_reset_delay_rejects_missing_unit | | FR-022, FR-027 | 0.00 s |
| OK | test_set_delay_accepts_leading_star_and_units | | FR-022, FR-027 | 0.00 s |
| OK | test_set_delay_rejects_missing_unit | | FR-022, FR-027 | 0.00 s |
| OK | test_status_category_must_be_error_warning_or_info | | FR-022, FR-027 | 0.00 s |
| OK | test_title_must_not_equal_description | | FR-022, FR-027 | 0.00 s |
| OK | test_title_too_short | | FR-022, FR-027 | 0.00 s |
| OK | test_turbine_platforms_required_when_provided_empty | | FR-022, FR-027 | 0.00 s |
| OK | test_valid_data_has_no_errors | | FR-022, FR-027 | 0.00 s |

## tests/integration/test_audit_chain_and_ops.py — 27 tests, 27 passed

I-18 tamper-evident audit log, I-09 verified backups, I-21 readiness / metrics / log file.

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_backup_is_written_verified_and_recorded | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 1.21 s |
| OK | test_a_configured_key_makes_the_chain_keyed | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 1.16 s |
| OK | test_a_corrupt_backup_fails_verification | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 1.29 s |
| OK | test_a_forged_row_cannot_be_repaired_without_the_key | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 1.16 s |
| OK | test_a_metrics_token_admits_an_address_outside_the_networks | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 0.60 s |
| OK | test_a_removed_row_breaks_the_link | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 1.21 s |
| OK | test_a_restored_backup_runs_the_application | The restore drill from docs/14 §5 in miniature: back up, start an app on the copy, log in. | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 1.79 s |
| OK | test_a_tampered_audit_log_inside_a_backup_is_flagged | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 1.28 s |
| OK | test_an_edited_row_is_found_and_named | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 1.39 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_archive_exports_old_rows_with_a_hash_and_deletes_nothing | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 1.46 s |
| OK | test_archive_with_nothing_old_enough_does_nothing | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 1.22 s |
| OK | test_backup_needs_a_file_database | An in-memory database cannot be backed up; built explicitly so the test also holds when | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 0.17 s |
| OK | test_cli_audit_verify_fails_with_a_non_zero_exit_when_broken | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 1.46 s |
| OK | test_cli_backup_verify_and_audit_commands | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 1.28 s |
| OK | test_every_audit_row_is_chained_to_its_predecessor | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 2.66 s |
| OK | test_metrics_are_refused_outside_the_allowed_networks | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 0.63 s |
| OK | test_metrics_count_requests_and_hold_no_business_data | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 1.28 s |
| OK | test_only_the_newest_generations_are_kept | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 1.40 s |
| OK | test_ready_is_503_when_the_chain_is_broken_or_the_schema_outdated | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 1.55 s |
| OK | test_ready_needs_no_login_and_reveals_no_business_data | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 1.01 s |
| OK | test_ready_reports_ok_and_degraded_by_backup_state | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 1.93 s |
| OK | test_requests_are_logged_to_the_configured_file | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 0.17 s |
| OK | test_rows_without_hashes_before_the_chain_started_are_tolerated | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 1.00 s |
| OK | test_the_append_only_triggers_still_stop_edits_inside_the_application | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 1.19 s |
| OK | test_the_hash_is_reproducible_from_the_stored_fields | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 1.22 s |
| OK | test_truncating_the_tail_is_caught_by_the_anchor | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 1.27 s |
| OK | test_verify_tail_detects_recent_tampering_cheaply | | AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044 | 1.17 s |

## tests/integration/test_audit_coverage.py — 7 tests, 7 passed

FR-056, FR-070, FR-072, FR-073, NFR-006, TC-015, TC-015a: the audit trail

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_audit_log_entries_cannot_be_updated_or_deleted | FR-072: append-only at the database layer, not just by omission. | FR-072 | 0.87 s |
| OK | test_each_lifecycle_action_is_audited_exactly_once | TC-015: one entry per state-changing action across the whole lifecycle. | TC-015 | 2.31 s |
| OK | test_import_lifecycle_is_audited | | FR-056, FR-070, FR-072, FR-073, NFR-006, TC-015, TC-015a | 0.89 s |
| OK | test_sandbox_delete_keeps_full_snapshot_and_creation_entry | TC-015a / FR-056 / FR-073. | FR-056, FR-073, TC-015a | 1.21 s |
| OK | test_subgroup_creation_is_audited_as_subgroup_create | | FR-056, FR-070, FR-072, FR-073, NFR-006, TC-015, TC-015a | 0.73 s |
| OK | test_unexpected_import_failure_marks_job_failed_and_is_audited | | FR-056, FR-070, FR-072, FR-073, NFR-006, TC-015, TC-015a | 0.85 s |
| OK | test_user_and_auth_events_are_audited | | FR-056, FR-070, FR-072, FR-073, NFR-006, TC-015, TC-015a | 1.05 s |

## tests/integration/test_audit_log.py — 4 tests, 4 passed

FR-070-FR-071: audit log search, Administrator full access,

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_admin_sees_all_entries | | FR-070, FR-071 | 0.88 s |
| OK | test_engineer_cannot_view_audit_log | | FR-070, FR-071 | 0.76 s |
| OK | test_filter_by_entity_type | | FR-070, FR-071 | 0.83 s |
| OK | test_reviewer_only_sees_own_actions | | FR-070, FR-071 | 1.00 s |

## tests/integration/test_auth.py — 7 tests, 7 passed

FR-001-FR-004, SEC-001-SEC-004.

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_deactivated_user_is_immediately_rejected | | FR-001, FR-004, SEC-001, SEC-004 | 0.94 s |
| OK | test_five_failed_attempts_locks_the_account | The 5th consecutive failure both locks the account (SEC-004) and | SEC-004 | 0.98 s |
| OK | test_login_success_returns_user_and_sets_session | | FR-001, FR-004, SEC-001, SEC-004 | 0.98 s |
| OK | test_logout_invalidates_session | | FR-001, FR-004, SEC-001, SEC-004 | 0.73 s |
| OK | test_me_requires_login | | FR-001, FR-004, SEC-001, SEC-004 | 0.57 s |
| OK | test_password_reset_invalidates_existing_session | | FR-001, FR-004, SEC-001, SEC-004 | 0.73 s |
| OK | test_wrong_password_is_rejected | | FR-001, FR-004, SEC-001, SEC-004 | 0.62 s |

## tests/integration/test_baseline_and_packages.py — 16 tests, 16 passed

The baseline of a Release and the all-languages / all-access-levels package (docs/03-srs.md

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_hub_controller_group_exists_for_the_controller_filter | | BR-007, FR-045, FR-173, FR-174, TC-066 | 0.60 s |
| OK | test_a_package_can_be_narrowed_to_a_platform_and_a_controller | | BR-007, FR-045, FR-173, FR-174, TC-066 | 2.84 s |
| OK | test_a_package_holds_every_language_for_every_access_level | | BR-007, FR-045, FR-173, FR-174, TC-066 | 3.94 s |
| OK | test_a_package_needs_a_published_release_and_valid_choices | | BR-007, FR-045, FR-173, FR-174, TC-066 | 3.03 s |
| OK | test_a_release_can_still_be_exported_as_only_its_own_changes | | BR-007, FR-045, FR-173, FR-174, TC-066 | 3.17 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_an_unpublished_release_previews_what_its_manual_will_contain | | BR-007, FR-045, FR-173, FR-174, TC-066 | 3.22 s |
| OK | test_everybody_may_download_what_is_released_and_only_owner_and_admin_what_is_not | | BR-007, FR-045, FR-173, FR-174, TC-066 | 4.48 s |
| OK | test_the_baseline_narrows_to_a_controller | | BR-007, FR-045, FR-173, FR-174, TC-066 | 2.38 s |
| OK | test_the_baseline_narrows_to_a_platform | | BR-007, FR-045, FR-173, FR-174, TC-066 | 2.91 s |
| OK | test_the_baseline_of_a_release_holds_every_code_valid_at_that_release | | BR-007, FR-045, FR-173, FR-174, TC-066 | 2.92 s |
| OK | test_the_manifest_vouches_for_every_file_and_reports_the_coverage | | BR-007, FR-045, FR-173, FR-174, TC-066 | 3.29 s |
| OK | test_the_package_is_audited | | BR-007, FR-045, FR-173, FR-174, TC-066 | 3.37 s |
| OK | test_the_readiness_check_shows_what_is_still_missing | | BR-007, FR-045, FR-173, FR-174, TC-066 | 3.33 s |
| OK | test_the_release_page_offers_the_baseline_and_the_package | | BR-007, FR-045, FR-173, FR-174, TC-066 | 3.54 s |
| OK | test_the_summary_counts_the_baseline_by_controller_and_platform | | BR-007, FR-045, FR-173, FR-174, TC-066 | 3.19 s |
| OK | test_the_title_page_names_the_scope | | BR-007, FR-045, FR-173, FR-174, TC-066 | 3.46 s |

## tests/integration/test_bulk_workflow.py — 19 tests, 19 passed

Bulk work on a change set with a sample inspection (docs/03-srs.md FR-180): edit, submit, domain

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_bulk_domain_signoff_needs_a_sample_and_every_code_gets_its_own_signoff | | BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070 | 1.78 s |
| OK | test_a_decision_in_the_workbench_names_its_sample | | BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070 | 2.48 s |
| OK | test_a_defect_in_the_sample_blocks_the_decision_until_it_is_sent_back | | BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070 | 2.56 s |
| OK | test_a_defect_needs_a_note_and_a_verdict_must_be_ok_or_defect | | BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070 | 2.12 s |
| OK | test_a_domain_signoff_is_only_for_a_holder_of_the_domain | | BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070 | 1.43 s |
| OK | test_a_sample_cannot_be_borrowed_reused_or_kept_after_the_batch_changed | | BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070 | 3.07 s |
| OK | test_a_small_batch_needs_no_sample | | BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070 | 1.81 s |
| OK | test_a_whole_domain_signoff_can_be_done_in_the_workbench | | BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070 | 2.11 s |
| OK | test_bulk_edit_refuses_unknown_fields_and_other_peoples_drafts | | BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070 | 1.49 s |
| OK | test_bulk_edit_sets_the_same_value_on_every_draft_with_its_own_audit_entry | | BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070 | 1.77 s |
| OK | test_bulk_submit_sends_the_drafts_that_are_ready_and_reports_the_rest | | BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070 | 1.88 s |
| OK | test_the_bulk_screens_are_translated | | BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070 | 1.80 s |
| OK | test_the_dossier_puts_the_codes_the_samples_and_the_audit_in_one_workbook | | BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070 | 2.25 s |
| OK | test_the_person_who_wrote_a_code_is_not_part_of_the_batch_they_decide | | BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070 | 2.42 s |
| OK | test_the_readiness_names_what_stops_each_draft | | BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070 | 1.39 s |
| OK | test_the_sample_is_spread_over_the_subgroups_and_reproducible | | BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070 | 0.00 s |
| OK | test_the_size_of_the_sample_follows_the_settings | | BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070 | 1.08 s |
| OK | test_the_workbench_shows_the_batch_the_readiness_and_the_rule | | BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070 | 2.17 s |
| OK | test_viewers_see_nothing_of_the_bulk_workbench | | BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070 | 1.22 s |

## tests/integration/test_change_requests.py — 11 tests, 11 passed

FR-030-FR-039, FR-027-FR-029, BR-004/BR-005. Full submit -> dual-review ->

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_author_cannot_sign_off_own_cr | | BR-004, BR-005, FR-027, FR-029, FR-030, FR-039 | 1.36 s |
| OK | test_chief_engineer_cannot_be_a_prior_signer | | BR-004, BR-005, FR-027, FR-029, FR-030, FR-039 | 1.71 s |
| OK | test_domain_signoff_rejects_user_without_that_domain | | BR-004, BR-005, FR-027, FR-029, FR-030, FR-039 | 0.99 s |
| OK | test_dual_signoff_in_either_order_reaches_pending_approval | | BR-004, BR-005, FR-027, FR-029, FR-030, FR-039 | 1.81 s |
| OK | test_full_happy_path_to_approved | | BR-004, BR-005, FR-027, FR-029, FR-030, FR-039 | 1.96 s |
| OK | test_rejection_returns_to_draft_and_clears_signoffs | | BR-004, BR-005, FR-027, FR-029, FR-030, FR-039 | 1.30 s |
| OK | test_reviewer_rejection_without_comment_is_rejected | | BR-004, BR-005, FR-027, FR-029, FR-030, FR-039 | 1.28 s |
| OK | test_submit_blocked_until_domain_signoff_complete | | BR-004, BR-005, FR-027, FR-029, FR-030, FR-039 | 1.07 s |
| OK | test_withdraw_forbidden_for_non_author | | BR-004, BR-005, FR-027, FR-029, FR-030, FR-039 | 0.89 s |
| OK | test_withdraw_removes_abandoned_new_draft | | BR-004, BR-005, FR-027, FR-029, FR-030, FR-039 | 0.76 s |
| OK | test_withdraw_revision_type_restores_prior_current_revision | Regression test: withdrawing a REVISION-type CR must restore the | BR-004, BR-005, FR-027, FR-029, FR-030, FR-039 | 2.35 s |

## tests/integration/test_connections.py — 12 tests, 12 passed

How status codes are connected: suppression by a cause, preconditions and escalation, read from the

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_negated_active_flag_is_a_cause_and_a_plain_one_a_precondition | | FR-048o, FR-092, FR-179, TC-069 | 0.61 s |
| OK | test_a_viewer_does_not_see_the_connections | | FR-048o, FR-092, FR-179, TC-069 | 1.42 s |
| OK | test_an_imported_program_brings_its_cause_and_its_repeat_code | | FR-048o, FR-092, FR-179, TC-069 | 1.30 s |
| OK | test_escalation_targets_say_which_are_outside_the_catalogue | | FR-048o, FR-092, FR-179, TC-069 | 1.09 s |
| OK | test_one_code_shows_what_it_depends_on_and_what_depends_on_it | | FR-048o, FR-092, FR-179, TC-069 | 1.11 s |
| OK | test_the_api_returns_the_connections_of_a_code_and_of_the_catalogue | | FR-048o, FR-092, FR-179, TC-069 | 1.12 s |
| OK | test_the_code_page_and_the_overview_show_the_connections | | FR-048o, FR-092, FR-179, TC-069 | 1.61 s |
| OK | test_the_hierarchy_puts_the_cause_on_top_with_what_it_suppresses_below | | FR-048o, FR-092, FR-179, TC-069 | 1.07 s |
| OK | test_the_hub_and_main_controller_can_be_looked_at_separately | | FR-048o, FR-092, FR-179, TC-069 | 1.09 s |
| OK | test_the_menu_offers_the_connections_to_the_working_roles_only | | FR-048o, FR-092, FR-179, TC-069 | 1.73 s |
| OK | test_the_real_programs_name_their_cause_and_their_repeat_code | | FR-048o, FR-092, FR-179, TC-069 | 0.63 s |
| OK | test_what_looks_wrong_is_listed | | FR-048o, FR-092, FR-179, TC-069 | 1.51 s |

## tests/integration/test_content_integrity.py — 16 tests, 16 passed

I-01 optimistic concurrency, I-04 reference warnings and where-used, I-12 duplicate

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_expected_updated_at_is_enforced_too | | BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039 | 1.24 s |
| OK | test_next_available_reports_how_many_numbers_are_left | | BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039 | 1.90 s |
| OK | test_planned_dates_are_administrator_set_audited_and_warn_at_publish | | BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039 | 2.49 s |
| OK | test_platform_variant_copies_the_code_and_links_the_family | | BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039 | 2.08 s |
| OK | test_reference_to_a_deprecated_code_warns | | BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039 | 2.89 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_related_codes_are_validated_and_checked | | BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039 | 1.23 s |
| OK | test_remedy_text_length_is_validated | | BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039 | 1.15 s |
| OK | test_save_without_a_version_still_works_for_old_clients | | BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039 | 1.56 s |
| OK | test_similar_endpoint_supports_the_did_you_mean_hint | | BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039 | 1.07 s |
| OK | test_similar_title_in_the_subgroup_is_flagged_not_blocked | | BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039 | 1.22 s |
| OK | test_stale_save_is_refused_with_the_current_values | | BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039 | 1.37 s |
| OK | test_troubleshooting_fields_round_trip_and_reach_the_export | | BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039 | 2.52 s |
| OK | test_ui_stale_edit_keeps_the_users_entries_and_shows_the_conflict | | BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039 | 1.67 s |
| OK | test_unknown_escalation_number_warns_but_does_not_block | | BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039 | 1.16 s |
| OK | test_variant_needs_a_decided_source_and_an_engineer | | BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039 | 1.30 s |
| OK | test_where_used_lists_dependants_and_deprecation_needs_acknowledgement | | BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039 | 2.54 s |

## tests/integration/test_corpus_report.py — 5 tests, 5 passed

The corpus report (I-68): what would the import do with every program under a folder, and why not.

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_program_the_mapping_does_not_know_is_one_line_of_the_reasons | | FR-048o, FR-192, FR-193, FR-201, TC-083 | 0.74 s |
| OK | test_every_program_of_a_folder_is_checked_and_nothing_is_created | | FR-048o, FR-192, FR-193, FR-201, TC-083 | 0.78 s |
| OK | test_the_command_prints_the_summary_and_writes_the_report | | FR-048o, FR-192, FR-193, FR-201, TC-083 | 0.78 s |
| OK | test_the_numbers_and_names_are_taken_out_of_a_reason | | FR-048o, FR-192, FR-193, FR-201, TC-083 | 0.63 s |
| OK | test_the_report_is_a_workbook_with_one_row_per_program | | FR-048o, FR-192, FR-193, FR-201, TC-083 | 1.21 s |

## tests/integration/test_csrf.py — 2 tests, 2 passed

docs/11-security-architecture.md §5: CSRF synchronizer-token protection

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_state_changing_request_with_csrf_token_succeeds | | | 0.60 s |
| OK | test_state_changing_request_without_csrf_token_is_rejected | | | 0.65 s |

## tests/integration/test_decision_trail.py — 5 tests, 5 passed

Where review comments live: the change-request page, the status-code

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_audit_entries_carry_the_comment_text_and_id | | | 1.54 s |
| OK | test_audit_log_can_be_filtered_by_action_and_actor | | | 1.79 s |
| OK | test_status_code_page_lists_all_decisions_and_comments | | | 2.41 s |
| OK | test_viewers_cannot_see_the_review_discussion | | | 1.56 s |
| OK | test_withdrawing_keeps_the_discussion_in_the_audit_log | | | 1.66 s |

## tests/integration/test_deprecation.py — 7 tests, 7 passed

FR-050-FR-052, BR-006, docs/05-governance-handbook.md §5, §11.

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_archive_blocked_before_retention_period_elapses | | BR-006, FR-050, FR-052 | 1.84 s |
| OK | test_archive_succeeds_after_retention_period | | BR-006, FR-050, FR-052 | 2.32 s |
| OK | test_deprecation_requires_released_state | | BR-006, FR-050, FR-052 | 0.80 s |
| OK | test_full_deprecation_cycle | | BR-006, FR-050, FR-052 | 2.57 s |
| OK | test_non_admin_cannot_archive_or_reinstate | | BR-006, FR-050, FR-052 | 0.79 s |
| OK | test_reinstate_deprecated_code | | BR-006, FR-050, FR-052 | 1.95 s |
| OK | test_second_revision_does_not_violate_current_uniqueness | Regression test: create_revision must flip the old current revision's | BR-006, FR-050, FR-052 | 2.35 s |

## tests/integration/test_email_and_windows_signin.py — 16 tests, 16 passed

The e-mail address of a user is fixed, and the Windows sign-in that will replace the password in a

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_domain_name_is_matched_through_the_mail_domain_or_the_user_name | | FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058 | 0.95 s |
| OK | test_a_header_from_an_address_that_is_not_a_trusted_proxy_is_not_believed | | FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058 | 0.90 s |
| OK | test_an_address_is_stored_trimmed_and_lower_case_and_must_be_valid_and_new | | FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058 | 1.24 s |
| OK | test_no_identity_an_unknown_one_and_a_deactivated_account_are_all_refused | | FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058 | 1.06 s |
| OK | test_production_never_starts_with_the_simulation_or_an_unknown_mode | | FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058 | 0.16 s |
| OK | test_the_address_of_a_user_cannot_be_changed | | FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058 | 1.11 s |
| OK | test_the_database_refuses_a_changed_address_too | | FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058 | 1.06 s |
| OK | test_the_development_defaults_are_the_simulation_next_to_the_password | | FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058 | 0.31 s |
| OK | test_the_gateways_identity_signs_the_matching_account_in_and_is_audited | | FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058 | 1.08 s |
| OK | test_the_screens_show_the_address_as_fixed | | FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058 | 1.35 s |
| OK | test_the_shipped_example_configuration_documents_the_settings | | FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058 | 0.02 s |
| OK | test_the_simulation_page_stands_in_for_the_gateway_and_signs_in_by_address | | FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058 | 1.03 s |
| OK | test_the_simulation_refuses_an_unknown_identity_and_is_off_by_default | | FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058 | 1.08 s |
| OK | test_with_both_modes_the_login_page_offers_the_button_and_the_form | | FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058 | 1.03 s |
| OK | test_with_passwords_only_the_windows_sign_in_is_not_there | | FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058 | 1.66 s |
| OK | test_with_windows_only_passwords_are_refused_and_the_login_page_says_so | | FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058 | 1.06 s |

## tests/integration/test_export_language.py — 4 tests, 4 passed

The words around an export follow the language of the export (docs/03-srs.md FR-171): the title

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_pdf_in_french_prints_its_own_words_in_french | | FR-045, FR-110, FR-165, FR-171, TC-065 | 0.11 s |
| OK | test_a_placeholder_survives_in_every_language | | FR-045, FR-110, FR-165, FR-171, TC-065 | 0.00 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_english_and_no_language_leave_the_text_as_it_is | | FR-045, FR-110, FR-165, FR-171, TC-065 | 0.09 s |
| OK | test_every_word_around_an_export_exists_in_every_language | | FR-045, FR-110, FR-165, FR-171, TC-065 | 0.00 s |

## tests/integration/test_exports_and_diff.py — 15 tests, 15 passed

FR-032, FR-045, FR-046, FR-054, FR-064, BR-008, SEC-023, TC-023, TC-035:

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_diff_card_renders_on_change_request_page | | BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035 | 2.09 s |
| OK | test_diff_hidden_from_viewers_for_open_revisions_and_unknown_revisions | | BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035 | 2.07 s |
| OK | test_export_file_names_carry_a_timestamp | Every export file name ends with the generation time, YYYY-MM-DD_HH-MM (UTC). | BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035 | 2.53 s |
| OK | test_exports_neutralise_formula_injection | | BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035 | 1.38 s |
| OK | test_json_export_is_flat_rows | | BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035 | 2.47 s |
| OK | test_new_code_diff_is_marked_new | | BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035 | 1.27 s |
| OK | test_release_can_be_deleted_before_publish_only | | BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035 | 2.35 s |
| OK | test_released_export_contains_only_released_non_sandbox_codes | | BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035 | 2.50 s |
| OK | test_released_export_filters_and_formats | | BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035 | 2.17 s |
| OK | test_revision_diff_endpoint_and_cr_baseline_diff | | BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035 | 2.19 s |
| OK | test_working_set_filters_by_lifecycle_state | | BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035 | 1.35 s |
| OK | test_working_set_is_labelled_and_excludes_sandbox | | BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035 | 1.46 s |
| OK | test_working_set_not_available_to_viewers_or_anonymous | | BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035 | 1.37 s |
| OK | test_working_set_pdf_and_xlsx_render | | BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035 | 1.95 s |
| OK | test_xlsx_export_has_same_rows_as_csv | TC-035. | TC-035 | 2.16 s |

## tests/integration/test_generated_fields.py — 20 tests, 20 passed

The fields of the generated controller programs (schema v4.1, docs/06-data-dictionary.md sections

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_field_of_the_other_controller_is_refused_and_the_rest_is_stored | | FR-025, FR-032, FR-048o, FR-168, TC-063 | 1.83 s |
| OK | test_a_hub_program_import_keeps_its_hub_fields_and_nothing_of_the_main_ones | | FR-025, FR-032, FR-048o, FR-168, TC-063 | 1.29 s |
| OK | test_a_main_program_import_keeps_every_field_of_the_program[StCd_1000.st] | | FR-025, FR-032, FR-048o, FR-168, TC-063 | 1.81 s |
| OK | test_a_main_program_import_keeps_every_field_of_the_program[StCd_1001.st] | | FR-025, FR-032, FR-048o, FR-168, TC-063 | 1.32 s |
| OK | test_a_main_program_import_keeps_every_field_of_the_program[StCd_1002.st] | | FR-025, FR-032, FR-048o, FR-168, TC-063 | 1.27 s |
| OK | test_a_named_audience_value_brings_its_default_levels_and_explicit_levels_win | | FR-025, FR-032, FR-048o, FR-168, TC-063 | 1.25 s |
| OK | test_a_new_revision_carries_over_what_the_caller_does_not_send_and_the_diff_shows_changes | | FR-025, FR-032, FR-048o, FR-168, TC-063 | 1.97 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_rejected_form_keeps_what_was_typed | | FR-025, FR-032, FR-048o, FR-168, TC-063 | 1.45 s |
| OK | test_the_csv_import_carries_the_new_columns_and_an_update_without_them_keeps_the_values | | FR-025, FR-032, FR-048o, FR-168, TC-063 | 1.29 s |
| OK | test_the_form_stores_the_fields_and_the_pages_show_them | | FR-025, FR-032, FR-048o, FR-168, TC-063 | 1.76 s |
| OK | test_the_levels_and_programs_are_internal_and_never_leave_in_an_audience_export | | FR-025, FR-032, FR-048o, FR-168, TC-063 | 2.43 s |
| OK | test_the_two_zero_load_attributes_are_not_mixed_up | | FR-025, FR-032, FR-048o, FR-168, TC-063 | 1.15 s |
| OK | test_values_out_of_range_are_refused[extra0] | | FR-025, FR-032, FR-048o, FR-168, TC-063 | 1.09 s |
| OK | test_values_out_of_range_are_refused[extra1] | | FR-025, FR-032, FR-048o, FR-168, TC-063 | 1.15 s |
| OK | test_values_out_of_range_are_refused[extra2] | | FR-025, FR-032, FR-048o, FR-168, TC-063 | 1.20 s |
| OK | test_values_out_of_range_are_refused[extra3] | | FR-025, FR-032, FR-048o, FR-168, TC-063 | 1.12 s |
| OK | test_values_out_of_range_are_refused[extra4] | | FR-025, FR-032, FR-048o, FR-168, TC-063 | 1.19 s |
| OK | test_values_out_of_range_are_refused[extra5] | | FR-025, FR-032, FR-048o, FR-168, TC-063 | 1.68 s |
| OK | test_values_out_of_range_are_refused[extra6] | | FR-025, FR-032, FR-048o, FR-168, TC-063 | 1.10 s |
| OK | test_withdrawing_and_deleting_a_code_removes_its_levels | | FR-025, FR-032, FR-048o, FR-168, TC-063 | 1.21 s |

## tests/integration/test_governance_extras.py — 24 tests, 24 passed

Remaining spec checks: SEC-010 service-layer authorization, FR-010/011/011a/014

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_user_holds_exactly_one_role | | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 0.80 s |
| OK | test_account_lock_and_password_events_are_audited | | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 1.74 s |
| OK | test_auth_provider_seam_is_substitutable | Callers depend on AuthenticatedUser only: a different provider returning | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 0.80 s |
| OK | test_comment_leaves_state_and_signoffs_untouched | | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 1.65 s |
| OK | test_import_preserves_legacy_reference_number | | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 1.26 s |
| OK | test_next_available_preview_does_not_reserve_a_number | | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 1.50 s |
| OK | test_services_enforce_roles_independently_of_route_decorators[-ADMINISTRATOR] | | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 0.61 s |
| OK | test_services_enforce_roles_independently_of_route_decorators[-CHIEF_ENGINEER0] | | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 0.64 s |
| OK | test_services_enforce_roles_independently_of_route_decorators[-CHIEF_ENGINEER1] | | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 0.56 s |
| OK | test_services_enforce_roles_independently_of_route_decorators[-ENGINEER0] | | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 0.60 s |
| OK | test_services_enforce_roles_independently_of_route_decorators[-ENGINEER1] | | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 0.60 s |
| OK | test_services_enforce_roles_independently_of_route_decorators[-ENGINEER2] | | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 0.58 s |
| OK | test_services_enforce_roles_independently_of_route_decorators[-ENGINEER3] | | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 0.61 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_services_enforce_roles_independently_of_route_decorators[-ENGINEER4] | | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 0.62 s |
| OK | test_services_enforce_roles_independently_of_route_decorators[-ENGINEER5] | | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 1.15 s |
| OK | test_services_enforce_roles_independently_of_route_decorators[-ENGINEER6] | | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 0.56 s |
| OK | test_services_enforce_roles_independently_of_route_decorators[-ENGINEER7] | | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 0.58 s |
| OK | test_services_enforce_roles_independently_of_route_decorators[-REVIEWER0] | | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 1.12 s |
| OK | test_services_enforce_roles_independently_of_route_decorators[-REVIEWER1] | | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 0.58 s |
| OK | test_services_enforce_roles_independently_of_route_decorators[-REVIEWER2] | | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 0.61 s |
| OK | test_services_enforce_roles_independently_of_route_decorators[-VIEWER0] | | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 0.53 s |
| OK | test_services_enforce_roles_independently_of_route_decorators[-VIEWER1] | | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 0.60 s |
| OK | test_user_administration_events_are_audited | | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 0.98 s |
| OK | test_whole_lifecycle_works_with_all_outbound_network_blocked | The application must keep operating when it can reach nothing beyond | FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010 | 2.81 s |

## tests/integration/test_import.py — 7 tests, 7 passed

FR-048a-d, EPIC-14: CSV/XLSX library import, still fully governed

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_ambiguous_group_code_requires_range_hint | WTUR is reused for a second, numerically distinct Hub Controller | FR-048a | 0.90 s |
| OK | test_imported_row_becomes_import_type_draft_change_request | | FR-048a | 0.86 s |
| OK | test_invalid_row_is_reported_and_creates_nothing | | FR-048a | 1.02 s |
| OK | test_non_admin_engineer_can_upload_import | | FR-048a | 0.76 s |
| OK | test_unknown_functional_system_group_code_reported_as_error | | FR-048a | 1.33 s |
| OK | test_unsupported_file_extension_rejected | | FR-048a | 0.78 s |
| OK | test_viewer_cannot_upload_import | | FR-048a | 0.74 s |

## tests/integration/test_import_toolkit.py — 20 tests, 20 passed

The migration toolkit around the library import (docs/21-improvement-ideas.md): I-05 template, dry

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_big_file_is_processed_in_the_background_with_progress | | BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043 | 2.70 s |
| OK | test_a_legacy_number_already_imported_is_refused_in_create_mode | | BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043 | 0.96 s |
| OK | test_dry_run_reports_every_problem_and_creates_nothing | | BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043 | 0.82 s |
| OK | test_dry_run_warns_about_a_possible_duplicate_without_blocking | | BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043 | 0.85 s |
| OK | test_error_report_has_row_column_value_and_a_fix_hint_as_csv_and_xlsx | | BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043 | 0.88 s |
| OK | test_files_above_the_threshold_go_to_the_background_automatically | | BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043 | 0.83 s |
| OK | test_formula_like_values_in_the_error_report_are_neutralised | | BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043 | 0.82 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_import_ui_pages_render_and_the_validate_button_creates_nothing | | BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043 | 1.73 s |
| OK | test_missing_required_columns_are_named_up_front | | BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043 | 1.29 s |
| OK | test_new_content_columns_are_imported | | BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043 | 1.29 s |
| OK | test_reconciliation_counts_states_and_names_the_missing_numbers | | BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043 | 2.17 s |
| OK | test_similar_titles_are_reported_as_warnings | | BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043 | 0.84 s |
| OK | test_template_csv_and_xlsx_carry_the_exact_headers_and_lookup_codes | | BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043 | 0.87 s |
| OK | test_the_downloaded_template_imports_cleanly_as_is | | BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043 | 0.87 s |
| OK | test_the_same_legacy_number_in_another_group_is_a_different_code | | BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043 | 0.88 s |
| OK | test_update_mode_errors_are_specific | | BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043 | 2.10 s |
| OK | test_update_mode_opens_a_new_revision_of_a_released_code | | BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043 | 2.16 s |
| OK | test_update_mode_revises_the_open_draft_in_place | | BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043 | 0.90 s |
| OK | test_update_mode_skips_unchanged_rows_and_creates_missing_ones | | BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043 | 0.81 s |
| OK | test_update_of_someone_elses_draft_is_reported_on_the_row | | BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043 | 1.06 s |

## tests/integration/test_input_hardening.py — 13 tests, 13 passed

What the API fuzz found (v4.18.1): an identifier wider than the database's integers is a missing

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_huge_identifier_in_a_body_is_not_a_server_error | | FR-200, NFR-006, SEC-020, TC-081 | 1.15 s |
| OK | test_a_password_may_hold_anything | | FR-200, NFR-006, SEC-020, TC-081 | 0.73 s |
| OK | test_a_text_with_control_characters_is_refused_everywhere | | FR-200, NFR-006, SEC-020, TC-081 | 1.07 s |
| OK | test_a_worksheet_never_gets_a_character_it_cannot_hold | | FR-200, NFR-006, SEC-020, TC-081 | 1.27 s |
| OK | test_an_identifier_wider_than_the_database_is_not_found[/api/v1/change-requests/1180591620717411303424] | | FR-200, NFR-006, SEC-020, TC-081 | 1.06 s |
| OK | test_an_identifier_wider_than_the_database_is_not_found[/api/v1/mapping-changes/1180591620717411303424] | | FR-200, NFR-006, SEC-020, TC-081 | 1.06 s |
| OK | test_an_identifier_wider_than_the_database_is_not_found[/api/v1/mapping/1180591620717411303424] | | FR-200, NFR-006, SEC-020, TC-081 | 1.16 s |
| OK | test_an_identifier_wider_than_the_database_is_not_found[/api/v1/packages/1180591620717411303424] | | FR-200, NFR-006, SEC-020, TC-081 | 1.15 s |
| OK | test_an_identifier_wider_than_the_database_is_not_found[/api/v1/releases/1180591620717411303424] | | FR-200, NFR-006, SEC-020, TC-081 | 1.17 s |
| OK | test_an_identifier_wider_than_the_database_is_not_found[/api/v1/status-codes/1180591620717411303424] | | FR-200, NFR-006, SEC-020, TC-081 | 1.15 s |
| OK | test_an_identifier_wider_than_the_database_is_not_found[/api/v1/users/11805916207174113034 24] | | FR-200, NFR-006, SEC-020, TC-081 | 1.19 s |
| OK | test_an_identifier_wider_than_the_database_is_not_found[/mapping/1180591620717411303424] | | FR-200, NFR-006, SEC-020, TC-081 | 1.09 s |
| OK | test_an_identifier_wider_than_the_database_is_not_found[/status-codes/1180591620717411303424] | | FR-200, NFR-006, SEC-020, TC-081 | 1.69 s |

## tests/integration/test_integrations_and_reports.py — 32 tests, 32 passed

I-29 API tokens, change feed and webhooks; I-26 saved searches; I-28 fleet occurrences;

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_disabled_webhook_is_not_called | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 2.56 s |
| OK | test_a_failing_webhook_is_recorded_and_never_blocks_the_publish | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 2.48 s |
| OK | test_a_hub_code_is_carried_over_as_a_main_controller_draft | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 1.90 s |
| OK | test_a_publish_calls_the_webhook_with_a_verifiable_signature | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 2.64 s |
| OK | test_a_token_is_shown_once_and_stored_only_as_a_hash | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 1.09 s |
| OK | test_a_token_only_works_for_its_scope | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 1.14 s |
| OK | test_a_token_reads_released_data_only_and_never_writes | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 2.84 s |
| OK | test_an_unreachable_webhook_target_does_not_block_the_publish | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 4.51 s |
| OK | test_bulk_deprecation_is_for_administrators_and_hub_groups | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 1.41 s |
| OK | test_carry_over_is_refused_for_the_wrong_kind_of_code_or_target | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 2.73 s |
| OK | test_health_report_covers_capacity_missing_data_and_inconsistencies | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 2.14 s |
| OK | test_health_report_finds_an_info_code_with_an_alarm | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 2.54 s |
| OK | test_health_report_flags_a_nearly_full_band | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 2.14 s |
| OK | test_hub_consolidation_ui_page_renders | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 2.05 s |
| OK | test_kpis_have_the_documented_shape | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 2.22 s |
| OK | test_occurrence_import_validation_and_permissions | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 1.29 s |
| OK | test_occurrence_report_names_codes_that_never_fire_and_those_that_fire_most | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 3.55 s |
| OK | test_redirects_are_not_followed | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 2.92 s |
| OK | test_reports_are_limited_to_the_right_roles | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 1.59 s |
| OK | test_revoked_expired_unknown_and_orphaned_tokens_are_refused | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 1.27 s |
| OK | test_sandbox_activity_never_appears_in_the_feed | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 1.19 s |
| OK | test_saved_search_ui_save_and_delete | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 1.36 s |
| OK | test_saved_searches_are_private_to_their_owner | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 1.30 s |
| OK | test_saved_searches_round_trip_through_the_url | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 1.42 s |
| OK | test_saving_under_an_existing_name_replaces_it_and_validates_the_name | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 1.17 s |
| OK | test_the_capacity_table_colours_main_blue_and_hub_light_blue | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 1.32 s |
| OK | test_the_feed_lists_releases_in_order_with_a_cursor_and_an_etag | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 3.90 s |
| OK | test_the_feed_needs_authentication | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 1.17 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_the_whole_consolidation_loop_ends_with_a_deprecation_request | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 3.74 s |
| OK | test_token_events_are_audited_without_the_secret | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 1.14 s |
| OK | test_token_management_is_for_administrators_and_validated | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 1.27 s |
| OK | test_webhook_targets_must_be_internal_and_inputs_valid | | FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046 | 1.28 s |

## tests/integration/test_invariants.py — 15 tests, 15 passed

Business-rule and data-integrity checks: BR-001..BR-003, BR-009, FR-023a,

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_draft_can_be_saved_repeatedly_without_workflow_transitions | | BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013 | 1.35 s |
| OK | test_duplicate_identifier_rejected_by_database | | BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013 | 1.38 s |
| OK | test_each_subgroup_sub_band_numbers_independently | | BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013 | 0.57 s |
| OK | test_exactly_one_current_revision_per_code | BR-002 / TC-011. | BR-002, TC-011 | 2.15 s |
| OK | test_filters_sort_and_pagination | | BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013 | 1.69 s |
| OK | test_full_text_search_matches_identifier_title_and_description | | BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013 | 1.63 s |
| OK | test_identifier_formats | | BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013 | 1.61 s |
| OK | test_identifier_never_changes_after_allocation | BR-001: a rejected-and-resubmitted CR keeps its identifier. | BR-001 | 1.77 s |
| OK | test_in_flight_revision_cannot_be_edited_in_place | FR-025: Review and later states are locked. | FR-025 | 1.48 s |
| OK | test_new_revision_of_a_released_code_keeps_its_identifier | BR-001 / FR-021: the identifier is allocated once, not again per revision. | BR-001, FR-021 | 2.41 s |
| OK | test_only_one_open_revision_per_code | BR-003 / TC-012: a second revision cannot be opened while one is open. | BR-003, TC-012 | 2.02 s |
| OK | test_released_code_stays_visible_to_viewers_while_a_new_revision_is_open | BR-009: opening revision 2 (a Draft) must not make the Released revision 1 | BR-009 | 2.39 s |
| OK | test_revision_history_visible_to_viewer_only_once_decided | FR-064 / BR-009. | BR-009, FR-064 | 2.55 s |
| OK | test_sequential_submits_in_two_subgroups_do_not_interfere | | BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013 | 2.70 s |
| OK | test_submit_requires_a_turbine_platform | | BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013 | 1.84 s |

## tests/integration/test_lookup_admin.py — 9 tests, 9 passed

US-091: Administrator manages controlled vocabularies. docs/05-governance-

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_admin_can_add_turbine_platform | | US-091 | 0.76 s |
| OK | test_admin_can_create_new_functional_system_group_in_free_range | FR-048g: into any currently free range (10000-10999, 14000-14999, | FR-048g | 0.79 s |
| OK | test_admin_can_deactivate_and_relabel_a_value | | US-091 | 0.81 s |
| OK | test_create_functional_subgroup_with_range | | US-091 | 0.79 s |
| OK | test_duplicate_code_rejected | | US-091 | 0.75 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_functional_subgroup_overlap_rejected | FR-048f: rejected with 409, naming the conflicting subgroup. | FR-048f | 0.71 s |
| OK | test_functional_system_group_not_creatable_via_generic_endpoint | v3 (EPIC-16): creation moved to a dedicated, non-overlap-validated | US-091 | 1.39 s |
| OK | test_new_functional_system_group_range_overlap_rejected | FR-048h: rejected with 409, naming the conflicting group. | FR-048h | 0.75 s |
| OK | test_non_admin_cannot_create_lookup_value | | US-091 | 0.73 s |

## tests/integration/test_mapping.py — 37 tests, 37 passed

The mapping of legacy to new status codes kept in the database and changed only with the three

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_bad_seed_workbook_is_refused_as_a_whole | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 0.70 s |
| OK | test_a_change_confirm_and_reject_work_on_stored_rows | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 2.21 s |
| OK | test_a_database_without_the_mapping_tables_says_what_to_run | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.93 s |
| OK | test_a_decision_asks_for_the_password_again | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.35 s |
| OK | test_a_new_row_takes_effect_only_after_all_three_approvals | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.63 s |
| OK | test_a_number_a_code_of_the_catalogue_carries_cannot_change | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.34 s |
| OK | test_a_number_held_by_the_catalogue_is_found_in_the_identifiers | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 0.73 s |
| OK | test_a_row_changed_in_between_makes_the_older_request_unusable | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 2.72 s |
| OK | test_a_row_that_does_not_pass_the_checks_makes_no_request[overrides0-description] | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.19 s |
| OK | test_a_row_that_does_not_pass_the_checks_makes_no_request[overrides1-type must be Error, Warning or Info] | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.16 s |
| OK | test_a_row_that_does_not_pass_the_checks_makes_no_request[overrides2-up/down counter must be Yes or No] | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.11 s |
| OK | test_a_row_that_does_not_pass_the_checks_makes_no_request[overrides3-new number must be a whole number] | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.07 s |
| OK | test_a_row_that_does_not_pass_the_checks_makes_no_request[overrides4-between 1 and 99999] | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.72 s |
| OK | test_a_row_that_does_not_pass_the_checks_makes_no_request[overrides5-neither a time] | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.11 s |
| OK | test_a_row_that_does_not_pass_the_checks_makes_no_request[overrides6-not a named level] | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.13 s |
| OK | test_a_row_that_does_not_pass_the_checks_makes_no_request[overrides7-belongs to the] | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.17 s |
| OK | test_a_row_that_does_not_pass_the_checks_makes_no_request[overrides8-cannot be approved without type] | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.20 s |
| OK | test_a_row_that_does_not_pass_the_checks_makes_no_request[overrides9-without logical node] | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.12 s |
| OK | test_a_second_administrator_cannot_replace_the_first_and_the_requester_cannot_decide | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 2.12 s |
| OK | test_a_viewer_neither_reads_nor_proposes | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.26 s |
| OK | test_a_withdrawn_import_is_blocked_when_its_mapping_row_is_gone | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.33 s |
| OK | test_a_workbook_makes_one_request_with_the_new_and_the_different_rows | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.57 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_workbook_that_unpacks_to_too_much_or_has_too_many_rows_is_refused | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.09 s |
| OK | test_a_workbook_with_a_wrong_row_names_it_and_requests_nothing | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.15 s |
| OK | test_numbers_stay_unique | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.25 s |
| OK | test_one_rejection_closes_the_request_and_needs_a_reason | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.47 s |
| OK | test_only_the_three_roles_decide_and_each_only_once | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.62 s |
| OK | test_show_all_puts_every_row_on_one_page_so_that_select_all_covers_them | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.24 s |
| OK | test_the_export_is_the_layout_the_upload_reads | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.20 s |
| OK | test_the_export_never_starts_a_formula | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.16 s |
| OK | test_the_menu_counts_the_requests_that_wait_for_a_decision | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.46 s |
| OK | test_the_pages_make_requests_and_decide_with_the_form_posts | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.61 s |
| OK | test_the_pages_show_the_mapping_the_requests_and_the_three_slots | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 2.31 s |
| OK | test_the_requester_can_withdraw_and_nobody_else | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.39 s |
| OK | test_the_seed_workbook_is_consistent | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 2.68 s |
| OK | test_the_seed_workbook_loads_into_an_empty_table_once | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 2.03 s |
| OK | test_the_sign_off_of_a_domain_is_below_the_changes_and_the_status_above | | BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079 | 1.32 s |

## tests/integration/test_menu.py — 22 tests, 22 passed

The menu on the left (docs/10-ui-ux-specification.md section 3, docs/03-srs.md FR-169): what each role

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_review_removes_the_mark_and_a_changed_string_brings_it_back | | FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064 | 1.64 s |
| OK | test_a_viewer_cannot_choose_or_keep_a_language_nobody_has_reviewed | | FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064 | 1.54 s |
| OK | test_a_viewer_sees_only_the_reviewed_languages_and_nothing_marked | | FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064 | 2.41 s |
| OK | test_a_viewer_sees_the_catalogue_and_nothing_to_work_with | | FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064 | 1.22 s |
| OK | test_a_viewer_who_reviews_a_language_sees_that_one | | FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064 | 1.26 s |
| OK | test_each_role_sees_the_links_of_its_work_and_no_other[admin] | | FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064 | 1.18 s |
| OK | test_each_role_sees_the_links_of_its_work_and_no_other[chief_engineer] | | FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064 | 1.35 s |
| OK | test_each_role_sees_the_links_of_its_work_and_no_other[engineer] | | FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064 | 1.23 s |
| OK | test_each_role_sees_the_links_of_its_work_and_no_other[reviewer] | | FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064 | 1.24 s |
| OK | test_each_role_sees_the_links_of_its_work_and_no_other[stcd_owner] | | FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064 | 1.80 s |
| OK | test_groups_can_be_closed_and_a_narrow_window_has_a_menu_button | | FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064 | 1.39 s |
| OK | test_no_link_appears_twice_and_the_groups_are_the_planned_ones | | FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064 | 1.25 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_only_the_link_of_the_current_page_is_marked | | FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064 | 1.94 s |
| OK | test_the_login_page_offers_only_the_reviewed_languages | | FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064 | 1.47 s |
| OK | test_the_marker_can_be_switched_off_in_the_settings | | FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064 | 1.87 s |
| OK | test_the_menu_is_translated_and_a_group_title_too | | FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064 | 1.30 s |
| OK | test_the_working_roles_see_every_language_and_the_unreviewed_ones_are_marked_beta[admin] | | FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064 | 1.38 s |
| OK | test_the_working_roles_see_every_language_and_the_unreviewed_ones_are_marked_beta[chief_engineer] | | FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064 | 1.32 s |
| OK | test_the_working_roles_see_every_language_and_the_unreviewed_ones_are_marked_beta[engineer] | | FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064 | 1.29 s |
| OK | test_the_working_roles_see_every_language_and_the_unreviewed_ones_are_marked_beta[reviewer] | | FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064 | 2.00 s |
| OK | test_the_working_roles_see_every_language_and_the_unreviewed_ones_are_marked_beta[stcd_owner] | | FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064 | 1.26 s |
| OK | test_what_waits_for_the_user_is_counted_in_the_menu | | FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064 | 1.68 s |

## tests/integration/test_multi_select.py — 7 tests, 7 passed

A filter that has several values takes several at once: a panel of tick boxes instead of a single

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_search_with_several_values_is_saved_and_comes_back | | FR-026, FR-196, FR-199, TC-080 | 1.35 s |
| OK | test_the_helpers_read_lists_repeats_and_comma_joined_strings | | FR-026, FR-196, FR-199, TC-080 | 0.00 s |
| OK | test_the_legacy_mapping_page_and_api_take_several_values | | FR-026, FR-196, FR-199, TC-080 | 1.87 s |
| OK | test_the_page_shows_a_panel_of_tick_boxes_with_the_chosen_ones_ticked | | FR-026, FR-196, FR-199, TC-080 | 1.36 s |
| OK | test_the_requests_filter_takes_several_states | | FR-026, FR-196, FR-199, TC-080 | 1.37 s |
| OK | test_the_status_code_list_takes_several_lifecycle_states_and_categories | | FR-026, FR-196, FR-199, TC-080 | 1.59 s |
| OK | test_the_status_filter_of_the_mapping_takes_several_values | | FR-026, FR-196, FR-199, TC-080 | 1.31 s |

## tests/integration/test_numbering.py — 4 tests, 4 passed

docs/07-database-design.md §7a: atomic Functional Group/Subgroup allocation.

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_allocation_increments_and_persists | | | 0.59 s |
| OK | test_capacity_exhaustion_raises | | | 0.69 s |
| OK | test_first_allocation_in_a_group_starts_at_range_start | | | 0.62 s |
| OK | test_sandbox_allocation_never_collides_with_real_range | | | 0.59 s |

## tests/integration/test_pdf_scripts.py — 20 tests, 20 passed

The PDF export prints every script of the languages the tool offers (docs/03-srs.md FR-165,

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_large_export_stays_fast_for_latin_text | | FR-045, FR-157, FR-165, NFR-008, TC-060 | 11.96 s |
| OK | test_a_plain_latin_export_loads_only_noto_sans | | FR-045, FR-157, FR-165, NFR-008, TC-060 | 0.05 s |
| OK | test_a_script_gets_its_own_font[arabic-\u0631\u0645\u0632 \u0627\u0644\u062d\u0627\u0644\u0629 \u0648\u062d\u062f\u0629 \u0627\u0644\u062a\u062d\u0643\u0645 StCd-01000] | | FR-045, FR-157, FR-165, NFR-008, TC-060 | 0.00 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_script_gets_its_own_font[bengali-\u09b8\u09cd\u09a5\u09bf\u09a4\u09bf \u0995\u09cb\u09a1 \u09a8\u09bf\u09af\u09bc\u09a8\u09cd\u09a4\u09cd\u09b0\u0995 \u09aa\u09b0\u09c0\u0995\u09cd\u09b7\u09be] | | FR-045, FR-157, FR-165, NFR-008, TC-060 | 0.00 s |
| OK | test_a_script_gets_its_own_font[devanagari-\u0938\u094d\u0925\u093f\u0924\u093f \u0915\u094b\u0921 \u0928\u093f\u092f\u0902\u0924\u094d\u0930\u0915 \u0915\u0940 \u091c\u093e\u0901\u091a] | | FR-045, FR-157, FR-165, NFR-008, TC-060 | 0.01 s |
| OK | test_a_script_gets_its_own_font[georgian-\u10e1\u10e2\u10d0\u10e2 \u10e3\u10e1\u10d8\u10e1 \u10d9\u10dd\u10d3\u10d8 \u10d9\u10dd\u10dc\u10e2\u10e0\u10dd\u10da\u10d4\u10e0\u10d8\u10e1 \u10e8\u10d4\u10db\u10dd\u10ec\u10db\u10d4\u10d1\u10d0] | | FR-045, FR-157, FR-165, NFR-008, TC-060 | 0.01 s |
| OK | test_a_script_gets_its_own_font[gujarati-\u0ab8\u0acd\u0aa5\u0abf\u0aa4\u0abf \u0a95\u0acb\u0aa1 \u0aa8\u0abf\u0aaf\u0a82\u0aa4\u0acd\u0ab0\u0a95 \u0aa4\u0aaa\u0abe\u0ab8] | | FR-045, FR-157, FR-165, NFR-008, TC-060 | 0.00 s |
| OK | test_a_script_gets_its_own_font[gurmukhi-\u0a38\u0a25\u0a3f\u0a24 \u0a40 \u0a15\u0a4b\u0a21 \u0a15\u0a70\u0a1f\u0a30\u0a4b\u0a32\u0a30 \u0a1c\u0a3e\u0a02\u0a1a] | | FR-045, FR-157, FR-165, NFR-008, TC-060 | 0.00 s |
| OK | test_a_script_gets_its_own_font[kannada-\u0cb8\u0ccd\u0ca5\u0cbf\u0ca4\u0cbf \u0c95\u0ccb\u0ca1\u0ccd \u0ca8\u0cbf\u0caf\u0c82\u0ca4\u0ccd\u0cb0\u0c95 \u0caa\u0cb0\u0cbf\u0cb6\u0cc0\u0cb2\u0ca8\u0cc6] | | FR-045, FR-157, FR-165, NFR-008, TC-060 | 0.00 s |
| OK | test_a_script_gets_its_own_font[malayalam-\u0d38\u0d4d\u0d25\u0d3f\u0d24\u0d3f \u0d15\u0d4b\u0d21\u0d4d \u0d28\u0d3f\u0d2f\u0d28\u0d4d\u0d24\u0d4d\u0d30\u0d15\u0d02 \u0d2a\u0d30\u0d3f\u0d36\u0d4b\u0d27\u0d28] | | FR-045, FR-157, FR-165, NFR-008, TC-060 | 0.00 s |
| OK | test_a_script_gets_its_own_font[odia-\u0b38\u0b4d\u0b25\u0b3f\u0b24\u0b3f \u0b15\u0b4b\u0b21\u0b4d \u0b28\u0b3f\u0b5f\u0b28\u0b4d\u0b24\u0b4d\u0b30\u0b15 \u0b2f\u0b3e\u0b1e\u0b4d\u0b1a] | | FR-045, FR-157, FR-165, NFR-008, TC-060 | 0.00 s |
| OK | test_a_script_gets_its_own_font[tamil-\u0ba8\u0bbf\u0bb2\u0bc8 \u0b95\u0bc1\u0bb1\u0bbf\u0baf\u0bc0\u0b9f\u0bc1 \u0b95\u0b9f\u0bcd\u0b9f\u0bc1\u0baa\u0bcd\u0baa\u0b9f\u0bc1\u0ba4\u0bcd\u0ba4\u0bbf \u0b9a\u0bcb\u0ba4\u0ba9\u0bc8] | | FR-045, FR-157, FR-165, NFR-008, TC-060 | 0.00 s |
| OK | test_a_script_gets_its_own_font[telugu-\u0c38\u0c4d\u0c25\u0c3f\u0c24\u0c3f \u0c15\u0c4b\u0c21\u0c4d \u0c28\u0c3f\u0c2f\u0c02\u0c24\u0c4d\u0c30\u0c3f\u0c15 \u0c24\u0c28\u0c3f\u0c16\u0c40] | | FR-045, FR-157, FR-165, NFR-008, TC-060 | 0.00 s |
| OK | test_a_script_gets_its_own_font[urdu-\u0627\u0633\u0679\u06cc\u0679\u0633 \u06a9\u0648\u0688 \u06a9\u0646\u0679\u0631\u0648\u0644\u0631 \u06a9\u06cc \u062c\u0627\u0646\u0686] | | FR-045, FR-157, FR-165, NFR-008, TC-060 | 0.00 s |
| OK | test_every_character_of_every_translation_is_drawn_by_a_bundled_font | | FR-045, FR-157, FR-165, NFR-008, TC-060 | 0.00 s |
| OK | test_every_script_builds_a_pdf_with_its_font | | FR-045, FR-157, FR-165, NFR-008, TC-060 | 0.58 s |
| OK | test_latin_text_needs_no_script_font_and_arabic_is_told_from_urdu | | FR-045, FR-157, FR-165, NFR-008, TC-060 | 0.00 s |
| OK | test_shaping_is_switched_on_per_cell_only_where_a_script_needs_it | | FR-045, FR-157, FR-165, NFR-008, TC-060 | 0.08 s |
| OK | test_the_fonts_and_their_licence_are_bundled_and_shaping_is_available | | FR-045, FR-157, FR-165, NFR-008, TC-060 | 0.04 s |
| OK | test_the_title_page_and_notice_may_carry_any_script | | FR-045, FR-157, FR-165, NFR-008, TC-060 | 0.18 s |

## tests/integration/test_prg_idx_and_user_groups.py — 9 tests, 9 passed

_Schema v4.1 (migration c9e2a4b6d815): PrgIdx_GUi on the status code, the controller-specific_

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_default_levels_are_seeded_only_where_the_programs_agree | | | 0.61 s |
| OK | test_leaving_draft_without_prg_idx_is_blocked_only_once_the_switch_is_on | | | 0.61 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_new_revision_columns_start_at_zero_or_not_captured | | | 1.24 s |
| OK | test_prg_idx_is_an_optional_uint_that_is_unique_among_real_codes | | | 0.61 s |
| OK | test_prg_idx_is_frozen_once_a_revision_was_released | | | 0.60 s |
| OK | test_the_locked_revision_trigger_covers_the_new_content_columns | | | 0.57 s |
| OK | test_the_migration_is_reversible_and_re_applies_cleanly | | | 1.51 s |
| OK | test_the_sixteen_user_group_slots_map_slots_one_to_eight_to_the_audiences | | | 0.60 s |
| OK | test_user_group_levels_are_bytes_and_unique_per_revision_and_slot | | | 0.61 s |

## tests/integration/test_publication.py — 25 tests, 25 passed

Publishing the catalogue (docs/21-improvement-ideas.md): I-02 profiles and wrapped PDF, I-03

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_administrator_manages_profiles_as_data | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 1.28 s |
| OK | test_applicability_matrix_and_changes_between_versions | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 3.75 s |
| OK | test_audience_export_only_contains_codes_the_audience_may_see | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 3.51 s |
| OK | test_bundle_contains_the_file_and_a_manifest_that_matches | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 2.20 s |
| OK | test_covert_visibility_is_data_an_administrator_can_flip | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 2.93 s |
| OK | test_customer_profile_applies_its_default_audience_and_working_set_refuses_audiences | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 3.00 s |
| OK | test_draft_source_text_cannot_be_translated_yet | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 1.28 s |
| OK | test_export_history_lists_hashes_per_release | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 2.26 s |
| OK | test_export_profile_fields_match_the_export_row | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 2.41 s |
| OK | test_offline_bundle_is_one_self_contained_page_with_the_data | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 2.89 s |
| OK | test_offline_bundle_respects_audiences_and_needs_a_published_release | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 2.87 s |
| OK | test_pdf_cells_are_wrapped_not_truncated | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 2.25 s |
| OK | test_pdf_column_widths_fit_the_page | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 0.64 s |
| OK | test_pdf_survives_characters_no_font_has | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 0.67 s |
| OK | test_publication_ui_pages_render | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 2.89 s |
| OK | test_release_notes_list_added_changed_and_deprecated | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 5.90 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_release_targets_limit_the_candidates_and_freeze_at_publish | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 2.92 s |
| OK | test_signed_manifest_detects_edits_to_the_manifest_itself | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 2.80 s |
| OK | test_three_profiles_are_seeded_and_shape_the_columns | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 2.07 s |
| OK | test_translation_validation_worklist_and_carry_over | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 2.99 s |
| OK | test_translation_workflow_with_light_review_and_language_exports | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 4.18 s |
| OK | test_unpublished_release_notes_are_an_administrator_preview | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 2.11 s |
| OK | test_verify_export_cli_works_without_the_database | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 2.16 s |
| OK | test_verify_recognises_an_untouched_file_and_a_tampered_one | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 2.32 s |
| OK | test_version_range_must_be_ordered_and_known | | BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042 | 1.28 s |

## tests/integration/test_releases.py — 6 tests, 6 passed

FR-040-FR-048, BR-006-BR-008.

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_cannot_change_items_after_publish | | BR-006, BR-008, FR-040, FR-048 | 1.83 s |
| OK | test_csv_export_of_a_release | | BR-006, BR-008, FR-040, FR-048 | 2.01 s |
| OK | test_full_database_export_requires_administrator | | BR-006, BR-008, FR-040, FR-048 | 0.97 s |
| OK | test_full_release_lifecycle | | BR-006, BR-008, FR-040, FR-048 | 2.73 s |
| OK | test_non_admin_cannot_create_release | | BR-006, BR-008, FR-040, FR-048 | 1.31 s |
| OK | test_sandbox_revision_rejected_from_release_items | | BR-006, BR-008, FR-040, FR-048 | 0.88 s |

## tests/integration/test_review_aids.py — 17 tests, 17 passed

I-23 field comments and review checklists, I-13 evidence attachments, I-11 change sets,

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_attachment_permissions_and_removal_rules | | FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040 | 1.99 s |
| OK | test_attachment_type_size_and_content_checks | | FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040 | 2.10 s |
| OK | test_author_attaches_evidence_and_reviewers_download_it | | FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040 | 1.61 s |
| OK | test_auto_assignment_on_submit_is_a_setting | | FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040 | 1.41 s |
| OK | test_bulk_decision_needs_the_password_once_and_a_wrong_one_counts_once | | FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040 | 2.64 s |
| OK | test_bulk_decision_skips_what_the_actor_may_not_decide | | FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040 | 1.99 s |
| OK | test_change_set_groups_requests_and_bulk_decisions_stay_individual | | FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040 | 1.98 s |
| OK | test_change_set_rules | | FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040 | 1.92 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_checklist_is_administrator_managed_and_inactive_items_are_not_required | | FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040 | 1.28 s |
| OK | test_checklist_must_be_confirmed_to_approve_and_is_recorded | | FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040 | 1.76 s |
| OK | test_field_comment_rules | | FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040 | 1.74 s |
| OK | test_least_loaded_reviewer_is_suggested_and_assigned | | FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040 | 1.69 s |
| OK | test_only_the_commenter_author_or_admin_can_resolve | | FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040 | 1.75 s |
| OK | test_out_of_office_reviewers_are_not_auto_assigned | | FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040 | 1.51 s |
| OK | test_rejection_does_not_need_the_checklist | | FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040 | 1.62 s |
| OK | test_removing_a_draft_attachment_deletes_the_file_and_withdrawal_purges_all | | FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040 | 1.38 s |
| OK | test_reviewer_comments_on_a_field_and_the_author_sees_and_resolves_it | | FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040 | 2.75 s |

## tests/integration/test_roles_and_languages.py — 40 tests, 40 passed

The StCd Owner role, the role colours, who may create users and hand out roles, and the three

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_string_missing_from_a_catalogue_shows_in_english | Every language is complete (FR-167), but a string that is ever missing must still show. | FR-167 | 1.29 s |
| OK | test_arabic_is_right_to_left_but_identifiers_codes_and_numbers_stay_as_they_are | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 1.42 s |
| OK | test_each_added_language_translates_the_login_page_and_falls_back_to_english[ar-\u062a\u0633\u062c\u064a\u0644 \u0627\u0644\u062f\u062e\u0648\u0644] | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 1.35 s |
| OK | test_each_added_language_translates_the_login_page_and_falls_back_to_english[bn-\u09b2\u0997 \u0987\u09a8] | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 0.69 s |
| OK | test_each_added_language_translates_the_login_page_and_falls_back_to_english[da-Log ind] | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 0.65 s |
| OK | test_each_added_language_translates_the_login_page_and_falls_back_to_english[es-Iniciar sesi\xf3n] | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 0.67 s |
| OK | test_each_added_language_translates_the_login_page_and_falls_back_to_english[fr-Se connecter] | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 0.74 s |
| OK | test_each_added_language_translates_the_login_page_and_falls_back_to_english[gu-\u0ab2\u0ac9\u0a97 \u0a87\u0aa8] | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 0.70 s |
| OK | test_each_added_language_translates_the_login_page_and_falls_back_to_english[hi-\u0932\u0949\u0917 \u0907\u0928 \u0915\u0930\u0947\u0902] | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 0.64 s |
| OK | test_each_added_language_translates_the_login_page_and_falls_back_to_english[it-Accedi] | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 0.72 s |
| OK | test_each_added_language_translates_the_login_page_and_falls_back_to_english[ka-\u10e8\u10d4\u10e1\u10d5\u10da \u10d0] | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 0.69 s |
| OK | test_each_added_language_translates_the_login_page_and_falls_back_to_english[kn-\u0cb2\u0cbe\u0c97\u0ccd \u0c87\u0ca8\u0ccd] | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 0.66 s |
| OK | test_each_added_language_translates_the_login_page_and_falls_back_to_english[ml-\u0d32\u0d4b\u0d17\u0d3f\u0d7b] | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 0.69 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_each_added_language_translates_the_login_page_and_falls_back_to_english[mr-\u0932\u0949\u0917 \u0907\u0928] | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 0.72 s |
| OK | test_each_added_language_translates_the_login_page_and_falls_back_to_english[or-\u0b32\u0b17\u0b4d \u0b07\u0b28\u0b4d] | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 0.67 s |
| OK | test_each_added_language_translates_the_login_page_and_falls_back_to_english[pa-\u0a32\u0a3e\u0a17 \u0a07\u0a28] | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 1.36 s |
| OK | test_each_added_language_translates_the_login_page_and_falls_back_to_english[ta-\u0b89\u0bb3\u0bcd\u0ba8\u0bc1\u0bb4\u0bc8] | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 0.66 s |
| OK | test_each_added_language_translates_the_login_page_and_falls_back_to_english[te-\u0c32\u0c3e\u0c17\u0c3f\u0c28\u0c4d] | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 0.69 s |
| OK | test_each_added_language_translates_the_login_page_and_falls_back_to_english[ur-\u0644\u0627\u06af \u0627\u0646] | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 0.70 s |
| OK | test_only_an_administrator_creates_users | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 1.52 s |
| OK | test_other_roles_cannot_open_the_user_list | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 1.33 s |
| OK | test_the_administrator_still_gives_every_role | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 1.14 s |
| OK | test_the_catalogues_agree_and_only_translate_real_strings | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 2.98 s |
| OK | test_the_choice_of_an_added_language_is_remembered_for_the_user | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 1.37 s |
| OK | test_the_language_names_have_the_country_letters_left_and_the_flag_right | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 0.62 s |
| OK | test_the_last_administrator_is_still_protected | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 1.82 s |
| OK | test_the_levels_order_the_roles_and_red_is_the_highest | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 0.63 s |
| OK | test_the_menu_shows_letters_left_and_a_drawn_flag_right_for_every_language | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 1.17 s |
| OK | test_the_stcd_owner_cannot_reach_the_other_roles[admin-VIEWER] | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 1.18 s |
| OK | test_the_stcd_owner_cannot_reach_the_other_roles[chief_engineer-REVIEWER] | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 1.14 s |
| OK | test_the_stcd_owner_cannot_reach_the_other_roles[engineer-ADMINISTRATOR] | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 1.15 s |
| OK | test_the_stcd_owner_cannot_reach_the_other_roles[reviewer-STCD_OWNER] | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 1.18 s |
| OK | test_the_stcd_owner_cannot_reach_the_other_roles[stcd_owner-VIEWER] | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 1.21 s |
| OK | test_the_stcd_owner_cannot_reach_the_other_roles[viewer-CHIEF_ENGINEER] | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 1.17 s |
| OK | test_the_stcd_owner_changes_nothing_but_the_role | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 1.19 s |
| OK | test_the_stcd_owner_moves_users_between_viewer_engineer_and_reviewer | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 1.94 s |
| OK | test_the_stcd_owner_reads_the_workflow_but_changes_nothing_in_it | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 1.42 s |
| OK | test_the_stcd_owner_sees_the_list_without_the_administrator_forms | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 1.37 s |
| OK | test_the_user_page_colours_every_role_and_red_is_the_administrator | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 1.16 s |
| OK | test_who_may_change_which_role_to_which | | BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054 | 0.64 s |

## tests/integration/test_schema_migration.py — 14 tests, 14 passed

US-003/US-004 acceptance criteria: fresh DB matches docs/artifacts/schema.sql

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_all_documented_tables_exist | | US-003, US-004 | 0.64 s |
| OK | test_first_subgroup_of_every_main_group_covers_the_first_200_numbers | Rule of 2026-10-22: e.g. Tower 5000-5199, Generator 2000-2199; the next | US-003, US-004 | 0.63 s |
| OK | test_foreign_keys_enforced | SEC/BR data-integrity depends on FK enforcement being on for every | US-003, US-004 | 0.64 s |
| OK | test_functional_system_group_code_is_reused_for_hub_controller | WTUR and WROT are each intentionally reused for a second, | US-003, US-004 | 0.58 s |
| OK | test_hub_sub_bands_are_aligned_to_the_hundreds_of_their_prefix | docs/05-governance-handbook.md section 2: 123xx owns 12300-12399; a band shorter than a | US-003, US-004 | 0.60 s |
| OK | test_nineteen_languages_seeded_with_country_letters_left_and_the_flag_right | | US-003, US-004 | 0.64 s |
| OK | test_no_placeholder_group_is_seeded | Only the 12 real groups exist out of the box; free ranges stay free. | US-003, US-004 | 0.64 s |
| OK | test_only_sandbox_status_codes_can_be_deleted | Exercises trg_prevent_nonsandbox_delete from docs/artifacts/schema.sql. | US-003, US-004 | 1.31 s |
| OK | test_second_migration_is_reversible_and_idempotent | A database already at the earlier baseline is brought up to date by the | US-003, US-004 | 1.98 s |
| OK | test_six_roles_seeded | | US-003, US-004 | 0.61 s |
| OK | test_subgroup_codes_have_no_dash_and_no_letter_after_the_x | | US-003, US-004 | 0.62 s |
| OK | test_thirty_eight_functional_subgroups_seeded | | US-003, US-004 | 0.61 s |
| OK | test_three_turbine_platforms_seeded | | US-003, US-004 | 0.63 s |
| OK | test_twelve_functional_system_groups_seeded | 9 Main Controller groups + 3 Hub Controller groups (WTUR/WROT/WPPD | US-003, US-004 | 0.61 s |

## tests/integration/test_schema_upgrade.py — 16 tests, 16 passed

Upgrading a database that was created by the previous release (Alembic d6f8b0c2e4a3, schema v3)

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_second_upgrade_is_a_no_op_and_the_downgrade_returns_to_v3_with_the_data | | AUD-074, BR-016, FR-147, FR-148, TC-045 | 2.61 s |
| OK | test_an_upgraded_database_has_the_same_shape_as_a_fresh_one | | AUD-074, BR-016, FR-147, FR-148, TC-045 | 1.90 s |
| OK | test_covert_is_not_visible_and_the_other_audiences_still_are | | AUD-074, BR-016, FR-147, FR-148, TC-045 | 1.94 s |
| OK | test_every_trigger_of_a_fresh_database_exists_after_the_upgrade | | AUD-074, BR-016, FR-147, FR-148, TC-045 | 1.82 s |
| OK | test_existing_data_survives_the_upgrade | | AUD-074, BR-016, FR-147, FR-148, TC-045 | 1.34 s |
| OK | test_hub_groups_become_hub_controller_groups | | AUD-074, BR-016, FR-147, FR-148, TC-045 | 1.32 s |
| OK | test_hub_sub_bands_are_shifted_to_the_hundreds_by_the_upgrade | The v3 database still has the legacy x001 bands; the upgrade aligns them. | AUD-074, BR-016, FR-147, FR-148, TC-045 | 1.30 s |
| OK | test_import_job_is_rebuilt_for_programs_without_losing_rows_or_references | v4.2: the CHECK of import_job.source_format has to widen to 'ST', which SQLite can only do by | AUD-074, BR-016, FR-147, FR-148, TC-045 | 17.59 s |
| OK | test_new_columns_default_sensibly_on_old_rows | | AUD-074, BR-016, FR-147, FR-148, TC-045 | 1.26 s |
| OK | test_old_audit_rows_are_chained_and_the_chain_verifies | | AUD-074, BR-016, FR-147, FR-148, TC-045 | 17.84 s |
| OK | test_the_languages_and_export_profiles_are_seeded | | AUD-074, BR-016, FR-147, FR-148, TC-045 | 1.29 s |
| OK | test_the_migrated_hash_equals_what_the_application_would_have_written | Migration and application share one canonical form (app/audit_hash.py). | AUD-074, BR-016, FR-147, FR-148, TC-045 | 1.27 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_the_search_index_is_filled_with_the_old_codes | | AUD-074, BR-016, FR-147, FR-148, TC-045 | 1.28 s |
| OK | test_the_subgroup_codes_lose_their_letters_and_dashes_in_the_upgrade | | AUD-074, BR-016, FR-147, FR-148, TC-045 | 1.10 s |
| OK | test_the_table_rebuilds_leave_every_foreign_key_intact | import_job (revision b3f5a7c9d124) and user (d4a6c8e0f236) are rebuilt to widen a CHECK; the | AUD-074, BR-016, FR-147, FR-148, TC-045 | 17.67 s |
| OK | test_the_upgraded_database_works_for_a_whole_workflow | | AUD-074, BR-016, FR-147, FR-148, TC-045 | 2.24 s |

## tests/integration/test_search_and_help.py — 16 tests, 16 passed

I-31 ranked, typo-tolerant search (with a substring fallback), I-35 in-app help content.

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_hit_in_the_title_outranks_a_hit_in_the_description | | FR-148, FR-154, TC-047 | 1.26 s |
| OK | test_a_new_code_and_an_edit_are_reflected_without_a_rebuild | | FR-148, FR-154, TC-047 | 1.41 s |
| OK | test_every_lifecycle_state_role_and_cr_state_has_help | | FR-148, FR-154, TC-047 | 0.01 s |
| OK | test_help_texts_are_plain_sentences | | FR-148, FR-154, TC-047 | 0.00 s |
| OK | test_new_terms_are_in_the_project_glossary | | FR-148, FR-154, TC-047 | 0.00 s |
| OK | test_prefix_and_multi_word_search | | FR-148, FR-154, TC-047 | 2.11 s |
| OK | test_search_ignores_fts_operators_and_punctuation | | FR-148, FR-154, TC-047 | 1.38 s |
| OK | test_several_misspelt_words_are_all_tolerated_together | Each word may expand to a group of close spellings; the groups must be joined with AND | FR-148, FR-154, TC-047 | 1.29 s |
| OK | test_state_badges_carry_their_glossary_text_as_a_tooltip | | FR-148, FR-154, TC-047 | 1.41 s |
| OK | test_the_change_request_screen_says_what_happens_next | | FR-148, FR-154, TC-047 | 1.65 s |
| OK | test_the_help_page_shows_the_quick_start_of_the_signed_in_role | | FR-148, FR-154, TC-047 | 1.43 s |
| OK | test_the_roles_and_states_are_named_in_the_project_glossary | | FR-148, FR-154, TC-047 | 0.00 s |
| OK | test_the_search_box_works_on_the_ui_list | | FR-148, FR-154, TC-047 | 1.53 s |
| OK | test_the_search_index_exists_on_a_fresh_database | | FR-148, FR-154, TC-047 | 17.16 s |
| OK | test_typos_still_find_the_code | | FR-148, FR-154, TC-047 | 1.37 s |
| OK | test_without_the_index_search_falls_back_to_substrings | | FR-148, FR-154, TC-047 | 2.29 s |

## tests/integration/test_select_all.py — 2 tests, 2 passed

A list with a checkbox per row has a checkbox in its header that ticks them all (the helper is

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_every_list_with_a_checkbox_per_row_can_be_selected_as_a_whole | | FR-159, FR-180, NFR-007, TC-036, TC-070 | 0.37 s |
| OK | test_the_header_box_and_the_row_boxes_share_their_class | | FR-159, FR-180, NFR-007, TC-036, TC-070 | 0.02 s |

## tests/integration/test_sla_and_notifications.py — 16 tests, 16 passed

I-08 SLA ages, stalled requests, in-app notices and reminders (docs/21-improvement-ideas.md),

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_assigned_reviewer_only_and_pending_approval_goes_to_chief_engineers | | FR-120, FR-121, FR-122, FR-123, TC-041 | 1.95 s |
| OK | test_business_days_skip_weekends | | FR-120, FR-121, FR-122, FR-123, TC-041 | 0.00 s |
| OK | test_kpis_are_computed_from_stored_timestamps | | FR-120, FR-121, FR-122, FR-123, TC-041 | 2.33 s |
| OK | test_kpis_flag_an_incomplete_audit_trail_and_an_ownerless_code | | FR-120, FR-121, FR-122, FR-123, TC-041 | 2.01 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_kpis_without_data_do_not_divide_by_zero | | FR-120, FR-121, FR-122, FR-123, TC-041 | 0.69 s |
| OK | test_levels_follow_the_thresholds | | FR-120, FR-121, FR-122, FR-123, TC-041 | 1.51 s |
| OK | test_mail_failure_never_breaks_the_reminder_job | | FR-120, FR-121, FR-122, FR-123, TC-041 | 1.50 s |
| OK | test_mark_all_read | | FR-120, FR-121, FR-122, FR-123, TC-041 | 1.63 s |
| OK | test_queues_show_the_waiting_time | | FR-120, FR-121, FR-122, FR-123, TC-041 | 1.49 s |
| OK | test_reminders_also_reach_a_covering_delegate_and_can_be_mailed | | FR-120, FR-121, FR-122, FR-123, TC-041 | 1.65 s |
| OK | test_reminders_go_to_whoever_holds_the_next_decision_once_a_day | | FR-120, FR-121, FR-122, FR-123, TC-041 | 1.78 s |
| OK | test_send_reminders_cli | | FR-120, FR-121, FR-122, FR-123, TC-041 | 0.62 s |
| OK | test_stalled_view_and_reassignment | | FR-120, FR-121, FR-122, FR-123, TC-041 | 1.71 s |
| OK | test_the_default_thresholds_are_the_documented_ones | | FR-120, FR-121, FR-122, FR-123, TC-041 | 0.66 s |
| OK | test_thresholds_are_configurable_and_validated | | FR-120, FR-121, FR-122, FR-123, TC-041 | 1.91 s |
| OK | test_withdrawn_cr_notices_are_removed_with_it | | FR-120, FR-121, FR-122, FR-123, TC-041 | 2.35 s |

## tests/integration/test_st_category_and_draft_delete.py — 7 tests, 7 passed

The status category of a .st import is the type of the mapping row (FR-181 superseded by FR-192:

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_category_chosen_in_the_request_changes_nothing | | FR-026, FR-159, FR-181, FR-182, FR-192, TC-053, TC-070 | 2.13 s |
| OK | test_a_row_without_a_type_blocks_the_program | | FR-026, FR-159, FR-181, FR-182, FR-192, TC-053, TC-070 | 1.22 s |
| OK | test_all_own_drafts_of_a_filter_are_deleted_on_every_page | | FR-026, FR-159, FR-181, FR-182, FR-192, TC-053, TC-070 | 1.77 s |
| OK | test_somebody_elses_drafts_cannot_be_ticked_or_deleted | | FR-026, FR-159, FR-181, FR-182, FR-192, TC-053, TC-070 | 1.62 s |
| OK | test_the_list_offers_tick_boxes_for_the_own_drafts_and_deletes_the_ticked | | FR-026, FR-159, FR-181, FR-182, FR-192, TC-053, TC-070 | 1.57 s |
| OK | test_the_pages_have_no_category_controls_any_more | | FR-026, FR-159, FR-181, FR-182, FR-192, TC-053, TC-070 | 1.62 s |
| OK | test_the_type_of_a_program_is_the_type_of_its_mapping_row | | FR-026, FR-159, FR-181, FR-182, FR-192, TC-053, TC-070 | 1.31 s |

## tests/integration/test_st_import.py — 29 tests, 29 passed

Status code import from the generated StCd.st programs and the legacy -> new mapping_

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_file_without_programs_or_too_large_is_refused_cleanly | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 0.62 s |
| OK | test_a_hub_program_keeps_its_legacy_number_and_gets_the_new_one_from_the_mapping | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 0.64 s |
| OK | test_a_legacy_hub_number_missing_from_the_mapping_leaves_no_identifier | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 0.63 s |
| OK | test_a_legacy_number_that_belongs_to_another_code_blocks_the_program | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 0.92 s |
| OK | test_a_number_defined_twice_in_one_upload_blocks_the_second | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 0.80 s |
| OK | test_a_preview_belongs_to_its_uploader_and_expires | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 1.02 s |
| OK | test_a_second_preview_shows_what_is_already_there_and_confirm_refuses_it | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 1.19 s |
| OK | test_a_used_prg_idx_is_a_warning_and_is_not_stored_twice | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 1.71 s |
| OK | test_a_zip_and_loose_files_are_read_alike_and_other_files_are_noted | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 0.80 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_zip_that_unpacks_too_large_is_refused | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 0.81 s |
| OK | test_allocation_skips_a_reserved_number_and_a_manual_override_cannot_take_it | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 1.00 s |
| OK | test_an_administrator_cannot_override_a_reserved_identifier | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 1.53 s |
| OK | test_an_expired_preview_is_refused_and_removed | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 0.85 s |
| OK | test_confirm_creates_drafts_that_reserve_their_numbers_in_one_change_set | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 1.11 s |
| OK | test_confirming_without_a_selection_takes_every_ready_program | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 0.92 s |
| OK | test_nothing_usable_is_a_clear_error | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 1.60 s |
| OK | test_only_engineers_and_administrators_import | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 0.79 s |
| OK | test_submitting_issues_exactly_the_reserved_number | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 1.45 s |
| OK | test_the_command_line_reads_the_hub_numbers_of_the_seed_workbook | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 0.68 s |
| OK | test_the_controller_can_be_forced_for_a_file | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 0.77 s |
| OK | test_the_database_refuses_a_second_reservation_of_a_number | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 0.95 s |
| OK | test_the_draft_carries_the_attributes_of_the_program_and_the_mapping | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 0.95 s |
| OK | test_the_mapping_exports_as_csv_and_xlsx | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 1.87 s |
| OK | test_the_mapping_is_not_for_viewers | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 0.78 s |
| OK | test_the_mapping_lists_legacy_to_new_and_counts_codes_without_a_legacy_number | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 1.18 s |
| OK | test_the_new_number_of_a_hub_program_is_the_one_of_its_mapping_row | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 0.75 s |
| OK | test_the_preview_lists_every_program_and_creates_nothing | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 1.05 s |
| OK | test_the_reader_takes_the_attributes_of_a_main_controller_program | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 1.34 s |
| OK | test_the_source_of_an_imported_code_can_be_read | | BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053 | 0.98 s |

## tests/integration/test_st_mapping_rules.py — 11 tests, 11 passed

What the .st import takes from the mapping and how it checks the program against the row (FR-192,

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_hub_row_for_a_main_program_is_not_taken | | BR-027, FR-189, FR-192, FR-193, TC-077 | 1.21 s |
| OK | test_a_program_already_in_the_catalogue_keeps_its_title_when_the_row_is_gone | | BR-027, FR-189, FR-192, FR-193, TC-077 | 1.37 s |
| OK | test_a_rejected_row_is_as_good_as_none | | BR-027, FR-189, FR-192, FR-193, TC-077 | 1.24 s |
| OK | test_a_row_that_is_not_approved_yet_is_used_with_a_warning | | BR-027, FR-189, FR-192, FR-193, TC-077 | 1.27 s |
| OK | test_a_row_without_cause_and_directive_leaves_the_remedy_text_empty | | BR-027, FR-189, FR-192, FR-193, TC-077 | 1.32 s |
| OK | test_a_time_in_the_program_wins_over_the_mapping | | BR-027, FR-189, FR-192, FR-193, TC-077 | 1.29 s |
| OK | test_a_title_that_does_not_say_what_the_name_says_is_flagged | | BR-027, FR-189, FR-192, FR-193, TC-077 | 1.20 s |
| OK | test_cause_and_directive_become_the_remedy_text | | BR-027, FR-189, FR-192, FR-193, TC-077 | 1.28 s |
| OK | test_the_glossary_is_a_check_not_a_translation | | BR-027, FR-189, FR-192, FR-193, TC-077 | 0.66 s |
| OK | test_the_legacy_number_of_the_header_must_be_the_one_of_the_row | | BR-027, FR-189, FR-192, FR-193, TC-077 | 1.15 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_the_mapping_gives_the_value_of_a_delay_the_program_sets_with_a_parameter | | BR-027, FR-189, FR-192, FR-193, TC-077 | 2.12 s |

## tests/integration/test_st_no_assumptions.py — 10 tests, 10 passed

The import assumes no value (FR-189): a value is read from the program, chosen by the user, or - when

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_delay_may_be_the_name_of_a_parameter_but_not_a_malformed_time | | BR-023, FR-048o, FR-189, FR-190, FR-192, TC-053 | 1.29 s |
| OK | test_a_delay_that_is_a_variable_stays_the_name_of_the_variable | | BR-023, FR-048o, FR-189, FR-190, FR-192, TC-053 | 2.10 s |
| OK | test_a_delay_the_program_does_not_have_says_so | | BR-023, FR-048o, FR-189, FR-190, FR-192, TC-053 | 1.33 s |
| OK | test_a_level_that_is_no_longer_active_is_not_guessed | | BR-023, FR-048o, FR-189, FR-190, FR-192, TC-053 | 1.33 s |
| OK | test_a_level_the_mapping_leaves_empty_stays_empty_and_is_listed | | BR-023, FR-048o, FR-189, FR-190, FR-192, TC-053 | 2.11 s |
| OK | test_a_level_whose_numbers_differ_from_the_program_is_named_and_warned_about | | BR-023, FR-048o, FR-189, FR-190, FR-192, TC-053 | 1.31 s |
| OK | test_a_program_that_gives_everything_lists_only_what_no_program_carries | | BR-023, FR-048o, FR-189, FR-190, FR-192, TC-053 | 1.34 s |
| OK | test_a_program_the_mapping_does_not_know_is_blocked_and_never_imported | | BR-023, FR-048o, FR-189, FR-190, FR-192, TC-053 | 1.30 s |
| OK | test_the_named_access_levels_come_from_the_mapping | | BR-023, FR-048o, FR-189, FR-190, FR-192, TC-053 | 1.23 s |
| OK | test_the_preview_warns_with_the_name_of_the_parameter | | BR-023, FR-048o, FR-189, FR-190, FR-192, TC-053 | 1.17 s |

## tests/integration/test_status_codes.py — 9 tests, 9 passed

FR-020-FR-029, FR-053-FR-056, FR-060-FR-064.

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_admin_sandbox_gets_identifier_immediately | | FR-020, FR-029, FR-053, FR-056, FR-060, FR-064 | 0.82 s |
| OK | test_create_rejects_invalid_fields | | FR-020, FR-029, FR-053, FR-056, FR-060, FR-064 | 0.83 s |
| OK | test_create_rejects_missing_functional_subgroup | BR-013: mandatory, never optional (corrected 2026-09-17). | BR-013 | 0.76 s |
| OK | test_engineer_can_create_draft_with_no_identifier | | FR-020, FR-029, FR-053, FR-056, FR-060, FR-064 | 0.78 s |
| OK | test_non_admin_cannot_create_sandbox | | FR-020, FR-029, FR-053, FR-056, FR-060, FR-064 | 1.58 s |
| OK | test_sandbox_delete_requires_admin_and_sandbox_flag | | FR-020, FR-029, FR-053, FR-056, FR-060, FR-064 | 1.08 s |
| OK | test_search_defaults_exclude_sandbox | | FR-020, FR-029, FR-053, FR-056, FR-060, FR-064 | 0.88 s |
| OK | test_viewer_cannot_create_status_code | | FR-020, FR-029, FR-053, FR-056, FR-060, FR-064 | 0.76 s |
| OK | test_viewer_cannot_see_draft_status_codes | | FR-020, FR-029, FR-053, FR-056, FR-060, FR-064 | 0.96 s |

## tests/integration/test_step_up_and_delegation.py — 16 tests, 16 passed

I-17 step-up re-authentication (SEC-026) and I-16 delegation / out-of-office cover.

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_admin_signoff_chief_decision_publish_and_full_export_all_need_the_password | | BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038 | 2.16 s |
| OK | test_correct_password_lets_the_signoff_through_and_is_audited | | BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038 | 2.31 s |
| OK | test_deactivated_delegator_ends_the_delegation | | BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038 | 1.66 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_delegate_can_sign_off_as_the_absent_reviewer_and_audit_shows_both | | BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038 | 1.74 s |
| OK | test_delegate_sees_the_review_queue_and_a_delegated_chief_engineer_can_approve | | BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038 | 2.62 s |
| OK | test_delegating_does_not_grant_other_privileges | | BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038 | 0.83 s |
| OK | test_delegation_does_not_lift_segregation_of_duties | The delegate acts as themselves: the author cannot sign their own CR through a delegation. | BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038 | 2.47 s |
| OK | test_delegation_is_limited_to_its_period_and_can_be_revoked | | BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038 | 1.79 s |
| OK | test_delegation_validation_and_who_may_create_it | | BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038 | 0.95 s |
| OK | test_one_person_cannot_give_both_review_signoffs_through_a_delegation | | BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038 | 1.78 s |
| OK | test_repeated_wrong_passwords_lock_the_account | The prompt must not be usable to guess a password (same lockout as login). | BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038 | 1.84 s |
| OK | test_reviewer_signoff_without_password_is_refused | | BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038 | 1.39 s |
| OK | test_step_up_can_be_switched_off_by_configuration | | BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038 | 1.50 s |
| OK | test_ui_decision_forms_carry_a_password_field | | BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038 | 1.77 s |
| OK | test_ui_signoff_without_password_shows_an_error | | BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038 | 1.78 s |
| OK | test_wrong_password_is_refused_and_audited | | BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038 | 1.57 s |

## tests/integration/test_terminology.py — 13 tests, 13 passed

The technical nomenclature: a vocabulary of its own, apart from the interface translations

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_formula_in_a_cell_is_neutralised_in_the_csv | | AUD-077, BR-018, FR-166, SEC-010, TC-061 | 1.37 s |
| OK | test_a_proposal_is_a_draft_until_another_person_approves_it | | AUD-077, BR-018, FR-166, SEC-010, TC-061 | 1.80 s |
| OK | test_a_shipped_draft_can_be_approved_by_anybody_who_may_approve_and_all_at_once | | AUD-077, BR-018, FR-166, SEC-010, TC-061 | 5.02 s |
| OK | test_an_interface_reviewer_has_no_say_in_the_terminology | | AUD-077, BR-018, FR-166, SEC-010, TC-061 | 1.23 s |
| OK | test_everybody_reads_but_only_the_technical_roles_propose | | AUD-077, BR-018, FR-166, SEC-010, TC-061 | 2.67 s |
| OK | test_only_the_managers_add_and_change_terms_and_a_new_english_reopens_approvals | | AUD-077, BR-018, FR-166, SEC-010, TC-061 | 2.47 s |
| OK | test_send_back_needs_a_note_and_reopens_an_approved_translation | | AUD-077, BR-018, FR-166, SEC-010, TC-061 | 5.67 s |
| OK | test_terms_that_are_never_translated_have_no_translation | | AUD-077, BR-018, FR-166, SEC-010, TC-061 | 0.63 s |
| OK | test_the_check_finds_interface_strings_that_do_not_use_the_term | | AUD-077, BR-018, FR-166, SEC-010, TC-061 | 2.19 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_the_shipped_terms_agree_with_the_interface_catalogue_where_both_have_the_word | Role names are interface strings and terms: they must be the same word. | AUD-077, BR-018, FR-166, SEC-010, TC-061 | 0.00 s |
| OK | test_the_vocabulary_is_exported_per_language_and_for_all | | AUD-077, BR-018, FR-166, SEC-010, TC-061 | 1.29 s |
| OK | test_the_vocabulary_ships_with_every_term_in_every_language_as_a_draft | | AUD-077, BR-018, FR-166, SEC-010, TC-061 | 1.47 s |
| OK | test_validation_of_a_translation | | AUD-077, BR-018, FR-166, SEC-010, TC-061 | 1.36 s |

## tests/integration/test_translation_approval.py — 12 tests, 12 passed

New and changed translations are approved separately, and every change of the text or of the

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_change_of_the_nomenclature_needs_the_flag_as_well | | FR-164, FR-170, FR-172, FR-176, TC-067 | 1.46 s |
| OK | test_a_changed_string_must_get_a_verdict_before_the_record_is_current | | FR-164, FR-170, FR-172, FR-176, TC-067 | 1.45 s |
| OK | test_a_delta_read_one_by_one_can_lower_or_keep_the_confidence | | FR-164, FR-170, FR-172, FR-176, TC-067 | 0.61 s |
| OK | test_a_follow_up_review_covers_only_what_is_new_or_changed | | FR-164, FR-170, FR-172, FR-176, TC-067 | 1.84 s |
| OK | test_a_language_never_reviewed_stays_hidden_from_a_viewer | | FR-164, FR-170, FR-172, FR-176, TC-067 | 1.27 s |
| OK | test_a_reviewed_language_has_nothing_pending_until_a_string_changes | | FR-164, FR-170, FR-172, FR-176, TC-067 | 2.15 s |
| OK | test_a_viewer_reads_a_changed_string_in_english_until_it_is_approved | | FR-164, FR-170, FR-172, FR-176, TC-067 | 1.72 s |
| OK | test_an_accepted_proposal_is_approved_and_an_undecided_one_is_not | | FR-164, FR-170, FR-172, FR-176, TC-067 | 2.06 s |
| OK | test_every_change_of_the_text_or_the_nomenclature_has_been_through_the_check | The gate: this fails while a string or a term changed and the check was not done for it. | FR-164, FR-170, FR-172, FR-176, TC-067 | 0.12 s |
| OK | test_the_colour_intervals_run_from_red_to_green | | FR-164, FR-170, FR-172, FR-176, TC-067 | 0.00 s |
| OK | test_the_overview_shows_the_confidence_of_every_language_in_colour | | FR-164, FR-170, FR-172, FR-176, TC-067 | 2.03 s |
| OK | test_the_record_holds_a_confidence_for_every_language_with_its_findings | | FR-164, FR-170, FR-172, FR-176, TC-067 | 0.01 s |

## tests/integration/test_translation_impact.py — 11 tests, 11 passed

A change of the vocabulary is tested automatically and its language impact reaches the StCd Owner

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_changed_nomenclature_is_named_in_the_notice | | FR-166, FR-176, FR-177, FR-178, TC-068 | 1.16 s |
| OK | test_a_changed_term_and_an_approved_translation_are_reported_again | | FR-166, FR-176, FR-177, FR-178, TC-068 | 1.22 s |
| OK | test_a_failing_notice_never_undoes_the_change | | FR-166, FR-176, FR-177, FR-178, TC-068 | 1.08 s |
| OK | test_a_new_term_tells_the_stcd_owner_which_languages_need_attention | | FR-166, FR-176, FR-177, FR-178, TC-068 | 1.12 s |
| OK | test_a_term_that_stays_in_english_touches_no_language | | FR-166, FR-176, FR-177, FR-178, TC-068 | 1.12 s |
| OK | test_changed_shipped_text_is_reported_once_per_change | | FR-166, FR-176, FR-177, FR-178, TC-068 | 1.12 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_nothing_is_reported_while_the_confidence_check_is_current | | FR-166, FR-176, FR-177, FR-178, TC-068 | 1.34 s |
| OK | test_the_command_for_the_deployment_reports_and_notifies | | FR-166, FR-176, FR-177, FR-178, TC-068 | 1.12 s |
| OK | test_the_notice_appears_on_the_notices_page_of_the_stcd_owner | | FR-166, FR-176, FR-177, FR-178, TC-068 | 2.06 s |
| OK | test_the_reviewer_of_an_affected_language_is_told_too | | FR-166, FR-176, FR-177, FR-178, TC-068 | 1.24 s |
| OK | test_the_start_of_the_server_and_the_installers_run_the_check | | FR-166, FR-176, FR-177, FR-178, TC-068 | 0.05 s |

## tests/integration/test_translation_review.py — 15 tests, 15 passed

Translation review in the application (docs/26-translations.md, docs/03-srs.md FR-164): native

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_missing_translation_can_be_proposed_and_accepted | | AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059 | 3.12 s |
| OK | test_a_proposal_awaits_approval_until_it_is_decided | | AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059 | 4.23 s |
| OK | test_a_proposal_must_keep_the_placeholders_and_an_unclear_string_is_only_noted | | AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059 | 6.42 s |
| OK | test_a_review_is_completed_only_when_every_string_has_a_verdict_and_is_recorded | | AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059 | 4.18 s |
| OK | test_a_reviewer_answers_strings_and_is_told_what_is_wrong | | AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059 | 3.04 s |
| OK | test_a_sheet_that_is_not_one_is_refused_clearly | | AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059 | 2.97 s |
| OK | test_an_accepted_proposal_is_shown_at_once_and_can_be_taken_back | | AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059 | 5.84 s |
| OK | test_an_account_that_reviews_has_history_in_the_usage_report | | AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059 | 2.75 s |
| OK | test_an_unassigned_user_cannot_open_a_review_and_an_unknown_language_is_404 | | AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059 | 1.24 s |
| OK | test_nobody_decides_on_their_own_proposal_and_a_decision_is_final | | AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059 | 7.34 s |
| OK | test_only_an_administrator_or_an_stcd_owner_assigns_reviewers_and_it_is_audited | | AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059 | 6.87 s |
| OK | test_right_to_left_languages_are_shown_right_to_left_in_the_review | | AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059 | 5.29 s |
| OK | test_the_overview_shows_a_reviewer_only_their_languages_and_a_manager_all | | AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059 | 5.46 s |
| OK | test_the_sheet_can_be_downloaded_filled_in_offline_and_uploaded | | AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059 | 4.19 s |
| OK | test_the_translations_page_does_not_scan_the_templates_once_per_language | The page needs the inventory of every string; it used to be rebuilt 37 times per request (6 s | AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059 | 2.40 s |

## tests/integration/test_ui_screens.py — 18 tests, 18 passed

S1–S19 server-rendered UI: every screen renders for the roles that can

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_admin_signoff_form_offers_access_override_and_submit_override | | NFR-007a, TC-036, UAT-13 | 1.95 s |
| OK | test_comments_api_validates_and_is_role_restricted | | NFR-007a, TC-036, UAT-13 | 1.48 s |
| OK | test_cr_detail_shows_role_specific_controls | | NFR-007a, TC-036, UAT-13 | 1.69 s |
| OK | test_default_role_for_new_user_is_viewer | SEC-012. | SEC-012 | 0.81 s |
| OK | test_draft_cr_page_says_it_must_still_be_submitted | | NFR-007a, TC-036, UAT-13 | 1.94 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_every_screen_renders_for_an_administrator | | NFR-007a, TC-036, UAT-13 | 3.37 s |
| OK | test_failed_form_submission_keeps_every_entered_value | NFR-007a / TC-036 / UAT-13. | NFR-007a, TC-036, UAT-13 | 1.44 s |
| OK | test_lookup_description_is_editable_and_flows_to_form_and_api | | NFR-007a, TC-036, UAT-13 | 1.68 s |
| OK | test_lookup_selects_carry_inline_help_hooks | NFR-007b: each lookup-backed select has a help target and per-option text. | NFR-007b | 1.41 s |
| OK | test_people_are_shown_by_name_and_lists_show_the_title | Authors, signers and commenters appear as "Full Name (Role)", never "User #n"; | NFR-007a, TC-036, UAT-13 | 3.17 s |
| OK | test_role_specific_screens_render[chief_engineer-/change-requests/approval-queue] | | NFR-007a, TC-036, UAT-13 | 1.30 s |
| OK | test_role_specific_screens_render[engineer-/change-requests/mine] | | NFR-007a, TC-036, UAT-13 | 1.25 s |
| OK | test_role_specific_screens_render[engineer-/exports] | | NFR-007a, TC-036, UAT-13 | 1.87 s |
| OK | test_role_specific_screens_render[reviewer-/change-requests/review-queue] | | NFR-007a, TC-036, UAT-13 | 1.28 s |
| OK | test_role_specific_screens_render[viewer-/exports] | | NFR-007a, TC-036, UAT-13 | 1.27 s |
| OK | test_role_specific_screens_render[viewer-/status-codes] | | NFR-007a, TC-036, UAT-13 | 1.34 s |
| OK | test_sign_off_request_listed_on_my_change_requests | | NFR-007a, TC-036, UAT-13 | 1.56 s |
| OK | test_ui_comment_reviewer_assignment_and_override_flow | | NFR-007a, TC-036, UAT-13 | 2.98 s |

## tests/integration/test_user_administration.py — 10 tests, 10 passed

Viewing, deactivating and deleting users (docs/03-srs.md FR-158, BR-025, SEC-032). An account that

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_deactivated_account_can_be_activated_again | | AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055 | 1.16 s |
| OK | test_an_account_that_was_never_used_can_be_deleted_and_the_deletion_is_audited | | AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055 | 1.42 s |
| OK | test_an_account_with_history_is_erased_and_the_audit_log_is_untouched | | AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055 | 1.52 s |
| OK | test_deleting_needs_the_administrators_password_and_an_administrator | | AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055 | 1.90 s |
| OK | test_nobody_deletes_their_own_account | | AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055 | 1.09 s |
| OK | test_the_buttons_deactivate_activate_and_delete_through_the_screens | | AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055 | 1.74 s |
| OK | test_the_detail_page_shows_history_and_the_buttons_follow_it | | AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055 | 2.61 s |
| OK | test_the_stcd_owner_sees_view_and_set_role_but_no_delete_or_deactivate | | AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055 | 1.41 s |
| OK | test_the_user_list_offers_view_edit_deactivate_and_delete_to_an_administrator | | AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055 | 1.34 s |
| OK | test_usage_is_open_to_the_user_list_roles_only | | AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055 | 1.40 s |

## tests/integration/test_users.py — 7 tests, 7 passed

FR-010-FR-014, SEC-003 §3 permission matrix, US-009.

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_admin_can_force_reset_another_users_password | | FR-010, FR-014, SEC-003, US-009 | 0.97 s |
| OK | test_admin_can_list_and_create_users | | FR-010, FR-014, SEC-003, US-009 | 0.81 s |
| OK | test_can_deactivate_administrator_when_another_remains_active | | FR-010, FR-014, SEC-003, US-009 | 0.86 s |
| OK | test_cannot_deactivate_last_active_administrator | | FR-010, FR-014, SEC-003, US-009 | 0.72 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_create_user_rejects_weak_password | | FR-010, FR-014, SEC-003, US-009 | 0.73 s |
| OK | test_non_admin_cannot_list_users | | FR-010, FR-014, SEC-003, US-009 | 0.81 s |
| OK | test_self_password_change | | FR-010, FR-014, SEC-003, US-009 | 1.14 s |

## tests/integration/test_withdraw_and_reimport.py — 11 tests, 11 passed

Withdrawing change requests (several at once) and what a withdrawal does to a program that was

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_change_request_of_somebody_else_is_refused_and_the_others_go_on | | BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056 | 2.37 s |
| OK | test_a_changed_program_opens_a_new_revision_of_an_approved_code | | BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056 | 2.27 s |
| OK | test_a_changed_program_updates_the_open_draft_when_it_is_ticked | | BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056 | 1.58 s |
| OK | test_a_restored_code_is_no_shell_and_the_database_refuses_a_delete_around_the_service | | BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056 | 2.70 s |
| OK | test_a_withdrawn_code_is_listed_and_an_administrator_deletes_it_and_imports_again | | BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056 | 2.19 s |
| OK | test_a_withdrawn_code_that_has_its_identifier_keeps_it_and_can_be_restored | | BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056 | 1.68 s |
| OK | test_a_withdrawn_import_draft_gives_its_number_and_legacy_number_back | | BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056 | 1.54 s |
| OK | test_bulk_withdraw_needs_a_list_and_a_role_that_can_withdraw | | BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056 | 1.65 s |
| OK | test_several_change_requests_are_withdrawn_in_one_go_and_each_is_reported | | BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056 | 1.30 s |
| OK | test_the_list_has_tick_boxes_and_a_withdraw_selected_button | | BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056 | 1.63 s |
| OK | test_the_preview_screen_lists_the_choices_and_leaves_changed_programs_unticked | | BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056 | 1.46 s |

## tests/integration/test_workflow_gaps.py — 18 tests, 18 passed

Closes the gaps found in the spec-vs-code review (docs/20-traceability-matrix.md):

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_admin_can_override_identifier_with_reason_and_it_is_audited | | BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013 | 1.51 s |
| OK | test_admin_signoff_can_override_proposed_audience_access | | BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013 | 1.84 s |
| OK | test_assigned_reviewer_only_may_sign_off | | BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013 | 1.89 s |
| OK | test_cr_page_explains_when_no_engineer_holds_a_pending_domain | | BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013 | 1.05 s |
| OK | test_domain_request_is_refused_when_nobody_holds_the_domain | A request nobody could ever act on must not be silently accepted. | BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013 | 0.93 s |
| OK | test_domain_signoff_request_surfaces_for_target_and_clears_once_signed | | BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013 | 1.63 s |
| OK | test_domain_signoff_request_validation | | BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013 | 1.25 s |
| OK | test_identifier_override_forbidden_for_engineer | | BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013 | 1.38 s |
| OK | test_identifier_override_rejects_out_of_range_malformed_and_duplicate | | BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013 | 3.03 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_identifier_override_requires_reason | | BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013 | 1.37 s |
| OK | test_no_reassignment_after_reviewer_signed_off | | BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013 | 2.58 s |
| OK | test_only_administrator_endpoint_accepts_overrides | | BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013 | 1.33 s |
| OK | test_rejection_ignores_access_overrides | | BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013 | 1.59 s |
| OK | test_reviewer_assignment_validation_and_authorization | | BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013 | 1.58 s |
| OK | test_reviewer_can_be_reassigned_and_unassigned | | BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013 | 1.85 s |
| OK | test_unknown_access_override_field_is_rejected | | BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013 | 1.53 s |
| OK | test_untargeted_domain_request_is_visible_to_every_domain_holder | | BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013 | 1.35 s |
| OK | test_withdrawn_cr_removes_its_domain_requests | | BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013 | 1.30 s |

## tests/system/test_accessibility.py — 9 tests, 9 passed

I-22 accessibility (WCAG 2.1 AA, docs/10-ui-ux-specification.md §12): colour contrast of the

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_every_page_is_structurally_accessible[de] | | FR-155, NFR-010, TC-049 | 3.67 s |
| OK | test_every_page_is_structurally_accessible[en] | | FR-155, NFR-010, TC-049 | 4.54 s |
| OK | test_every_state_badge_meets_aa | | FR-155, NFR-010, TC-049 | 0.00 s |
| OK | test_status_is_never_conveyed_by_colour_alone | Every state badge carries the state as text, not only a colour. | FR-155, NFR-010, TC-049 | 1.49 s |
| OK | test_text_tokens_meet_aa_on_their_backgrounds | | FR-155, NFR-010, TC-049 | 0.00 s |
| OK | test_the_chief_engineers_queue_is_accessible | | FR-155, NFR-010, TC-049 | 1.29 s |
| OK | test_the_contrast_helper_matches_the_wcag_reference_values | | FR-155, NFR-010, TC-049 | 0.01 s |
| OK | test_the_login_page_is_accessible | | FR-155, NFR-010, TC-049 | 0.68 s |
| OK | test_the_stylesheet_honours_user_preferences | | FR-155, NFR-010, TC-049 | 0.00 s |

## tests/system/test_api_fuzz.py — 173 tests, 170 passed

Property-based fuzz of the whole REST API against docs/artifacts/openapi.yaml (schemathesis): for

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_no_operation_answers_with_a_server_error | | FR-200, NFR-006, SEC-020, TC-082 | 25.24 s |
| OK | test_no_operation_answers_with_a_server_error[DELETE /api/v1/api-tokens/{tokenId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[DELETE /api/v1/attachments/{attachmentId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[DELETE /api/v1/change-sets/{changeSetId}/change-requests/{crId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[DELETE /api/v1/delegations/{delegationId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[DELETE /api/v1/releases/{releaseId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[DELETE /api/v1/saved-searches/{searchId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_no_operation_answers_with_a_server_error[DELETE /api/v1/status-codes/{statusCodeId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/api-tokens] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/attachments/{attachmentId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/audiences] | | FR-200, NFR-006, SEC-020, TC-082 | 0.05 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/audit-log] | | FR-200, NFR-006, SEC-020, TC-082 | 0.12 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/auth/csrf-token] | | FR-200, NFR-006, SEC-020, TC-082 | 0.01 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/auth/me] | | FR-200, NFR-006, SEC-020, TC-082 | 0.05 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/bulk-samples/{sampleId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/change-requests/stalled] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/change-requests/{crId}/attachments] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/change-requests/{crId}/domain-signoffs] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/change-requests/{crId}/field-comments] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/change-requests/{crId}/review-checklist] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/change-requests/{crId}/sla] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/change-requests/{crId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/change-requests] | | FR-200, NFR-006, SEC-020, TC-082 | 0.10 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/change-sets/{changeSetId}/dossier.xlsx] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/change-sets/{changeSetId}/readiness] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/change-sets/{changeSetId}/samples] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/change-sets/{changeSetId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/change-sets] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/changes] | | FR-200, NFR-006, SEC-020, TC-082 | 0.08 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/delegations] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/domain-signoff-requests] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/export-profiles] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/exports/released] | | FR-200, NFR-006, SEC-020, TC-082 | 0.21 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/exports/working-set] | | FR-200, NFR-006, SEC-020, TC-082 | 0.25 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/exports/{jobId}/manifest] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/health/ready] | | FR-200, NFR-006, SEC-020, TC-082 | 0.23 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/health] | | FR-200, NFR-006, SEC-020, TC-082 | 0.01 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/hub-consolidation/mapping] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/imports/reconciliation] | | FR-200, NFR-006, SEC-020, TC-082 | 0.08 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/imports/template] | | FR-200, NFR-006, SEC-020, TC-082 | 0.20 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/imports/{importJobId}/errors] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/imports/{importJobId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/imports] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/legacy-mapping/export] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/legacy-mapping] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/lookups/{vocabulary}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.08 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/mapping-changes/{changeId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/mapping-changes] | | FR-200, NFR-006, SEC-020, TC-082 | 0.08 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/mapping/export] | | FR-200, NFR-006, SEC-020, TC-082 | 1.43 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/mapping/{mappingId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/mapping] | | FR-200, NFR-006, SEC-020, TC-082 | 0.12 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/notifications] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/occurrence-imports] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/operations/backup-status] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/packages/{jobId}/download] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/packages/{jobId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/releases/{releaseId}/baseline] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/releases/{releaseId}/candidates] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/releases/{releaseId}/export] | | FR-200, NFR-006, SEC-020, TC-082 | 0.10 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/releases/{releaseId}/exports] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/releases/{releaseId}/notes] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/releases/{releaseId}/offline-bundle] | | FR-200, NFR-006, SEC-020, TC-082 | 0.08 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/releases/{releaseId}/packages] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/releases/{releaseId}/readiness] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/releases/{releaseId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.05 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/releases] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/reports/catalogue-health] | | FR-200, NFR-006, SEC-020, TC-082 | 0.12 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/reports/kpis] | | FR-200, NFR-006, SEC-020, TC-082 | 0.10 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/reports/occurrences] | | FR-200, NFR-006, SEC-020, TC-082 | 0.08 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/review-checklist-items] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/reviewers/workload] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/revisions/{revisionId}/translations] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/saved-searches] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/settings] | | FR-200, NFR-006, SEC-020, TC-082 | 0.08 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/software-versions/changes] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/software-versions/matrix] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/status-codes/connections] | | FR-200, NFR-006, SEC-020, TC-082 | 0.08 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/status-codes/next-available] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/status-codes/similar] | | FR-200, NFR-006, SEC-020, TC-082 | 0.09 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/status-codes/{statusCodeId}/connections] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/status-codes/{statusCodeId}/family] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/status-codes/{statusCodeId}/import-source] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/status-codes/{statusCodeId}/revisions/{revisionA}/diff/{revisionB}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/status-codes/{statusCodeId}/revisions] | | FR-200, NFR-006, SEC-020, TC-082 | 0.08 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/status-codes/{statusCodeId}/where-used] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/status-codes/{statusCodeId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.08 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/status-codes] | | FR-200, NFR-006, SEC-020, TC-082 | 0.38 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/translations/worklist] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/users/{userId}/usage] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/users/{userId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/users] | | FR-200, NFR-006, SEC-020, TC-082 | 2.35 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/webhooks/{webhookId}/deliveries] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[GET /api/v1/webhooks] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[PATCH /api/v1/export-profiles/{profileId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.20 s |
| OK | test_no_operation_answers_with_a_server_error[PATCH /api/v1/lookups/{vocabulary}/{valueId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[PATCH /api/v1/review-checklist-items/{itemId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.08 s |
| OK | test_no_operation_answers_with_a_server_error[PATCH /api/v1/revisions/{revisionId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.09 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_no_operation_answers_with_a_server_error[PATCH /api/v1/users/{userId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.08 s |
| OK | test_no_operation_answers_with_a_server_error[PATCH /api/v1/webhooks/{webhookId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.08 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/api-tokens] | | FR-200, NFR-006, SEC-020, TC-082 | 0.10 s |
| SKIP | test_no_operation_answers_with_a_server_error[POST /api/v1/auth/login] | | FR-200, NFR-006, SEC-020, TC-082 | 0.11 s |
| SKIP | test_no_operation_answers_with_a_server_error[POST /api/v1/auth/logout] | | FR-200, NFR-006, SEC-020, TC-082 | 0.02 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/auth/windows] | | FR-200, NFR-006, SEC-020, TC-082 | 0.02 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/bulk-samples/{sampleId}/reject-defects] | | FR-200, NFR-006, SEC-020, TC-082 | 0.08 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/change-requests/withdraw] | | FR-200, NFR-006, SEC-020, TC-082 | 0.12 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/change-requests/{crId}/admin-signoff] | | FR-200, NFR-006, SEC-020, TC-082 | 0.12 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/change-requests/{crId}/attachments] | | FR-200, NFR-006, SEC-020, TC-082 | 0.28 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/change-requests/{crId}/chief-engineer-decide] | | FR-200, NFR-006, SEC-020, TC-082 | 0.11 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/change-requests/{crId}/comments] | | FR-200, NFR-006, SEC-020, TC-082 | 0.11 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/change-requests/{crId}/domain-signoff-requests] | | FR-200, NFR-006, SEC-020, TC-082 | 1.04 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/change-requests/{crId}/domain-signoffs] | | FR-200, NFR-006, SEC-020, TC-082 | 0.32 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/change-requests/{crId}/field-comments] | | FR-200, NFR-006, SEC-020, TC-082 | 0.31 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/change-requests/{crId}/reviewer-assignment/auto] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/change-requests/{crId}/reviewer-signoff] | | FR-200, NFR-006, SEC-020, TC-082 | 0.11 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/change-requests/{crId}/submit] | | FR-200, NFR-006, SEC-020, TC-082 | 0.30 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/change-requests/{crId}/withdraw] | | FR-200, NFR-006, SEC-020, TC-082 | 0.08 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/change-sets/{changeSetId}/bulk-edit] | | FR-200, NFR-006, SEC-020, TC-082 | 0.16 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/change-sets/{changeSetId}/bulk-submit] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/change-sets/{changeSetId}/change-requests] | | FR-200, NFR-006, SEC-020, TC-082 | 0.13 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/change-sets/{changeSetId}/close] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/change-sets/{changeSetId}/decisions] | | FR-200, NFR-006, SEC-020, TC-082 | 0.16 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/change-sets/{changeSetId}/domain-signoffs] | | FR-200, NFR-006, SEC-020, TC-082 | 0.15 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/change-sets/{changeSetId}/samples] | | FR-200, NFR-006, SEC-020, TC-082 | 0.12 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/change-sets] | | FR-200, NFR-006, SEC-020, TC-082 | 0.20 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/delegations] | | FR-200, NFR-006, SEC-020, TC-082 | 0.15 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/export-profiles] | | FR-200, NFR-006, SEC-020, TC-082 | 0.17 s |
| SKIP | test_no_operation_answers_with_a_server_error[POST /api/v1/exports/full-database] | | FR-200, NFR-006, SEC-020, TC-082 | 0.03 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/exports/verify] | | FR-200, NFR-006, SEC-020, TC-082 | 0.12 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/field-comments/{commentId}/resolve] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/hub-consolidation/deprecation-requests] | | FR-200, NFR-006, SEC-020, TC-082 | 0.10 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/imports/reconciliation] | | FR-200, NFR-006, SEC-020, TC-082 | 0.14 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/imports/st/preview] | | FR-200, NFR-006, SEC-020, TC-082 | 0.15 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/imports/st] | | FR-200, NFR-006, SEC-020, TC-082 | 0.16 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/imports] | | FR-200, NFR-006, SEC-020, TC-082 | 0.19 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/lookups/functional-subgroups] | | FR-200, NFR-006, SEC-020, TC-082 | 0.32 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/lookups/functional-system-groups] | | FR-200, NFR-006, SEC-020, TC-082 | 0.33 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/lookups/{vocabulary}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.09 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/mapping-changes/workbook] | | FR-200, NFR-006, SEC-020, TC-082 | 0.18 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/mapping-changes/{changeId}/decisions] | | FR-200, NFR-006, SEC-020, TC-082 | 0.15 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/mapping-changes/{changeId}/withdraw] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/mapping-changes] | | FR-200, NFR-006, SEC-020, TC-082 | 0.44 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/mapping/seed] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/notifications/read-all] | | FR-200, NFR-006, SEC-020, TC-082 | 0.64 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/notifications/{notificationId}/read] | | FR-200, NFR-006, SEC-020, TC-082 | 0.06 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/occurrence-imports] | | FR-200, NFR-006, SEC-020, TC-082 | 0.20 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/operations/audit-verify] | | FR-200, NFR-006, SEC-020, TC-082 | 0.10 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/releases/{releaseId}/packages] | | FR-200, NFR-006, SEC-020, TC-082 | 0.12 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/releases/{releaseId}/publish] | | FR-200, NFR-006, SEC-020, TC-082 | 0.10 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/releases] | | FR-200, NFR-006, SEC-020, TC-082 | 0.12 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/review-checklist-items] | | FR-200, NFR-006, SEC-020, TC-082 | 0.17 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/revisions/{revisionId}/translations/carry-over] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/revisions/{revisionId}/translations/{language}/approve] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/saved-searches] | | FR-200, NFR-006, SEC-020, TC-082 | 0.16 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/status-codes/{statusCodeId}/archive] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/status-codes/{statusCodeId}/deprecation-requests] | | FR-200, NFR-006, SEC-020, TC-082 | 0.10 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/status-codes/{statusCodeId}/mc-drafts] | | FR-200, NFR-006, SEC-020, TC-082 | 0.10 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/status-codes/{statusCodeId}/platform-variants] | | FR-200, NFR-006, SEC-020, TC-082 | 0.31 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/status-codes/{statusCodeId}/reinstate] | | FR-200, NFR-006, SEC-020, TC-082 | 0.12 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/status-codes/{statusCodeId}/revisions] | | FR-200, NFR-006, SEC-020, TC-082 | 0.32 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/status-codes] | | FR-200, NFR-006, SEC-020, TC-082 | 0.39 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/users/me/change-password] | | FR-200, NFR-006, SEC-020, TC-082 | 0.09 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/users/{userId}/delete] | | FR-200, NFR-006, SEC-020, TC-082 | 0.11 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/users/{userId}/force-reset-password] | | FR-200, NFR-006, SEC-020, TC-082 | 0.12 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/users] | | FR-200, NFR-006, SEC-020, TC-082 | 0.24 s |
| OK | test_no_operation_answers_with_a_server_error[POST /api/v1/webhooks] | | FR-200, NFR-006, SEC-020, TC-082 | 0.11 s |
| OK | test_no_operation_answers_with_a_server_error[PUT /api/v1/bulk-samples/{sampleId}/items/{changeRequestId}] | | FR-200, NFR-006, SEC-020, TC-082 | 0.19 s |
| OK | test_no_operation_answers_with_a_server_error[PUT /api/v1/change-requests/{crId}/reviewer-assignment] | | FR-200, NFR-006, SEC-020, TC-082 | 0.10 s |
| OK | test_no_operation_answers_with_a_server_error[PUT /api/v1/releases/{releaseId}/items] | | FR-200, NFR-006, SEC-020, TC-082 | 0.07 s |
| OK | test_no_operation_answers_with_a_server_error[PUT /api/v1/releases/{releaseId}/target-versions] | | FR-200, NFR-006, SEC-020, TC-082 | 0.12 s |
| OK | test_no_operation_answers_with_a_server_error[PUT /api/v1/revisions/{revisionId}/translations/{language}] | | FR-200, NFR-006, SEC-020, TC-082 | 1.07 s |
| OK | test_no_operation_answers_with_a_server_error[PUT /api/v1/settings] | | FR-200, NFR-006, SEC-020, TC-082 | 0.35 s |
| OK | test_no_operation_answers_with_a_server_error[PUT /api/v1/status-codes/{statusCodeId}/planned-dates] | | FR-200, NFR-006, SEC-020, TC-082 | 0.15 s |
| OK | test_the_fuzz_is_really_signed_in | | FR-200, NFR-006, SEC-020, TC-082 | 5.99 s |

## tests/system/test_branding.py — 5 tests, 5 passed

The identity of the product (FR-187): the full name, the owner on top of every title page and

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_every_title_page_carries_the_identity | | FR-185, FR-186, FR-187, TC-074 | 0.30 s |
| OK | test_the_guide_names_the_same_owner | | FR-185, FR-186, FR-187, TC-074 | 0.04 s |
| OK | test_the_logo_files_exist_and_are_drawn | | FR-185, FR-186, FR-187, TC-074 | 0.23 s |
| OK | test_the_menu_bar_and_the_login_page_show_the_logo | | FR-185, FR-186, FR-187, TC-074 | 1.48 s |
| OK | test_the_owner_the_long_name_and_the_logo_are_written_once | | FR-185, FR-186, FR-187, TC-074 | 0.01 s |

## tests/system/test_cli_create_admin.py — 2 tests, 2 passed

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_create_admin_cli_command | | | 0.66 s |
| OK | test_create_admin_cli_rejects_weak_password | | | 0.56 s |

## tests/system/test_dark_mode.py — 8 tests, 8 passed

Dark mode (docs/10-ui-ux-specification.md section 13, FR-183): the dark palette meets the same

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_colours_are_only_written_out_where_the_dark_theme_cannot_matter | Outside the token blocks a hex colour is allowed only for fills that carry their own text | FR-155, FR-183, NFR-010, SEC-025, TC-049 | 0.01 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_every_badge_and_sla_pill_meets_aa_in_the_dark_theme | | FR-155, FR-183, NFR-010, SEC-025, TC-049 | 0.00 s |
| OK | test_the_dark_palette_meets_aa_on_its_backgrounds | | FR-155, FR-183, NFR-010, SEC-025, TC-049 | 0.01 s |
| OK | test_the_dark_theme_gives_every_light_tint_a_dark_value | A token added to the light palette without a dark twin would stay light on a dark page. | FR-155, FR-183, NFR-010, SEC-025, TC-049 | 0.00 s |
| OK | test_the_dark_theme_is_for_the_screen_only | | FR-155, FR-183, NFR-010, SEC-025, TC-049 | 0.00 s |
| OK | test_the_login_page_and_every_signed_in_page_offer_the_theme_choice | | FR-155, FR-183, NFR-010, SEC-025, TC-049 | 1.55 s |
| OK | test_the_theme_choice_is_translated | | FR-155, FR-183, NFR-010, SEC-025, TC-049 | 1.32 s |
| OK | test_theme_js_is_a_plain_file_that_follows_the_system_and_remembers_the_choice | | FR-155, FR-183, NFR-010, SEC-025, TC-049 | 0.01 s |

## tests/system/test_deploy_artifacts.py — 22 tests, 22 passed

I-33 / I-34: the packaging, installer, scheduler units and CI definitions are consistent with

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_backup_runs_every_night_and_reminders_on_weekdays | | NFR-013, NFR-014, TC-051 | 0.00 s |
| OK | test_ci_files_parse_and_run_the_ci_script | | NFR-013, NFR-014, TC-051 | 0.02 s |
| OK | test_create_admin_can_take_its_answers_from_the_environment | | NFR-013, NFR-014, TC-051 | 0.63 s |
| OK | test_each_timer_has_a_matching_oneshot_service | | NFR-013, NFR-014, TC-051 | 0.01 s |
| OK | test_every_expected_artifact_is_shipped | | NFR-013, NFR-014, TC-051 | 0.02 s |
| OK | test_every_pattern_in_the_scan_is_classified | | NFR-013, NFR-014, TC-051 | 0.00 s |
| OK | test_every_setting_an_installer_writes_is_read_by_the_application | | NFR-013, NFR-014, TC-051 | 0.03 s |
| OK | test_every_setting_in_the_example_file_is_read_by_the_application | | NFR-013, NFR-014, TC-051 | 0.01 s |
| OK | test_linux_installer_parses_and_its_dry_run_changes_nothing | | NFR-013, NFR-014, TC-051 | 3.10 s |
| OK | test_non_sqlite_connections_are_left_alone | The engine listeners must not touch another driver's connection (docs/07 section 12). | NFR-013, NFR-014, TC-051 | 0.68 s |
| OK | test_production_refuses_to_start_with_a_weak_key | | NFR-013, NFR-014, TC-051 | 9.57 s |
| OK | test_serve_options_follow_the_environment | | NFR-013, NFR-014, TC-051 | 0.05 s |
| OK | test_services_call_only_commands_that_exist | | NFR-013, NFR-014, TC-051 | 0.64 s |
| OK | test_the_ci_script_documents_itself | | NFR-013, NFR-014, TC-051 | 2.63 s |
| OK | test_the_postgres_ratchet_holds | | NFR-013, NFR-014, TC-051 | 11.74 s |
| OK | test_the_production_entry_point_serves_and_reports_its_state | | NFR-013, NFR-014, TC-051 | 20.07 s |
| OK | test_the_ratchet_notices_new_sqlite_specific_code | | NFR-013, NFR-014, TC-051 | 10.68 s |
| OK | test_units_parse_and_have_the_required_sections | | NFR-013, NFR-014, TC-051 | 0.04 s |
| OK | test_web_service_runs_the_production_entry_point_as_an_unprivileged_user | | NFR-013, NFR-014, TC-051 | 0.01 s |
| OK | test_windows_installer_dry_run_lists_every_step_and_creates_nothing | | NFR-013, NFR-014, TC-051 | 4.40 s |
| OK | test_windows_scripts_parse | | NFR-013, NFR-014, TC-051 | 0.86 s |
| OK | test_write_paths_of_the_units_cover_what_the_jobs_write | | NFR-013, NFR-014, TC-051 | 0.01 s |

## tests/system/test_docs_consistency.py — 17 tests, 17 passed

TC-052. The documents agree with the code and with each other: the generated endpoint table, the

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_every_requirement_id_cited_in_a_document_is_defined | | TC-052 | 0.70 s |
| OK | test_every_requirement_is_accounted_for_and_the_matrix_evidence_resolves | | TC-052 | 1.40 s |
| OK | test_every_screen_path_in_the_ui_inventory_is_a_real_route | docs/10 section 1 names the path of each v4 screen; each must exist as a UI route. | TC-052 | 1.20 s |
| OK | test_every_test_case_id_cited_in_the_documents_exists | | TC-052 | 0.16 s |
| OK | test_links_between_documents_point_at_existing_files_and_headings | | TC-052 | 0.37 s |
| OK | test_paragraphs_in_documents_are_single_source_lines | docs/15 §6: prose paragraphs are one continuous source line (tables, lists and code may span). | TC-052 | 0.12 s |
| OK | test_section_references_between_documents_exist | | TC-052 | 0.36 s |
| OK | test_the_audit_catalogue_lists_every_code_the_application_can_write | | TC-052 | 0.85 s |
| OK | test_the_contract_version_and_operation_count_quoted_in_the_documents | | TC-052 | 0.83 s |
| OK | test_the_decisions_and_assumptions_are_complete | | TC-052 | 0.05 s |
| OK | test_the_document_set_version_is_current_and_the_report_is_generated | | TC-052 | 0.01 s |
| OK | test_the_endpoint_table_of_the_api_document_is_the_generated_one | | TC-052 | 1.11 s |
| OK | test_the_group_and_subgroup_counts_quoted_in_the_documents | | TC-052 | 0.01 s |
| OK | test_the_screen_count_matches_the_inventory | | TC-052 | 0.03 s |
| OK | test_the_table_and_trigger_counts_quoted_in_the_documents | | TC-052 | 0.06 s |
| OK | test_the_test_suite_table_of_the_strategy_matches_the_test_files | docs/13 section 8a quotes how many test functions each v4 module has. | TC-052 | 0.21 s |
| OK | test_the_v4_tables_of_the_data_dictionary_name_real_columns | docs/06 section 13 lists each v4 table with its fields; every named field must exist. | TC-052 | 0.02 s |

## tests/system/test_docs_pdf.py — 36 tests, 36 passed

The documents of a release as PDF with the version in the file name (FR-185,

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_document_is_built_with_the_version_in_its_name | | FR-184, FR-185, NFR-013, TC-072 | 0.54 s |
| OK | test_an_unknown_document_is_refused | | FR-184, FR-185, NFR-013, TC-072 | 0.01 s |
| OK | test_every_character_of_the_documents_can_be_drawn[00-INDEX.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.01 s |
| OK | test_every_character_of_the_documents_can_be_drawn[01-executive-summary.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.01 s |
| OK | test_every_character_of_the_documents_can_be_drawn[02-prd.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.01 s |
| OK | test_every_character_of_the_documents_can_be_drawn[03-srs.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.04 s |
| OK | test_every_character_of_the_documents_can_be_drawn[04-domain-model.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.01 s |
| OK | test_every_character_of_the_documents_can_be_drawn[05-governance-handbook.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.01 s |
| OK | test_every_character_of_the_documents_can_be_drawn[06-data-dictionary.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.02 s |
| OK | test_every_character_of_the_documents_can_be_drawn[07-database-design.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.01 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_every_character_of_the_documents_can_be_drawn[08-system-architecture.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.01 s |
| OK | test_every_character_of_the_documents_can_be_drawn[09-api-specification.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.02 s |
| OK | test_every_character_of_the_documents_can_be_drawn[10-ui-ux-specification.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.02 s |
| OK | test_every_character_of_the_documents_can_be_drawn[11-security-architecture.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.01 s |
| OK | test_every_character_of_the_documents_can_be_drawn[12-audit-compliance.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.01 s |
| OK | test_every_character_of_the_documents_can_be_drawn[13-test-strategy.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.01 s |
| OK | test_every_character_of_the_documents_can_be_drawn[14-deployment-architecture.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.01 s |
| OK | test_every_character_of_the_documents_can_be_drawn[15-development-standards.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.01 s |
| OK | test_every_character_of_the_documents_can_be_drawn[16-mvp-roadmap.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.01 s |
| OK | test_every_character_of_the_documents_can_be_drawn[17-implementation-backlog.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.01 s |
| OK | test_every_character_of_the_documents_can_be_drawn[18-claude-code-implementation-guide.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.01 s |
| OK | test_every_character_of_the_documents_can_be_drawn[19-glossary.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.01 s |
| OK | test_every_character_of_the_documents_can_be_drawn[20-traceability-matrix.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.02 s |
| OK | test_every_character_of_the_documents_can_be_drawn[21-improvement-ideas.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.01 s |
| OK | test_every_character_of_the_documents_can_be_drawn[22-software-description.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.01 s |
| OK | test_every_character_of_the_documents_can_be_drawn[23-user-manual.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.02 s |
| OK | test_every_character_of_the_documents_can_be_drawn[24-test-report.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.06 s |
| OK | test_every_character_of_the_documents_can_be_drawn[25-hc-integration-concept.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.01 s |
| OK | test_every_character_of_the_documents_can_be_drawn[27-product-review-and-roadmap.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.02 s |
| OK | test_every_character_of_the_documents_can_be_drawn[28-operations-runbook.md] | A new symbol in a document must be given a word in _WORDS, or it prints as a gap. | FR-184, FR-185, NFR-013, TC-072 | 0.01 s |
| OK | test_inline_markup_is_removed_and_kept_as_styles | | FR-184, FR-185, NFR-013, TC-072 | 0.00 s |
| OK | test_symbols_a_pdf_font_cannot_draw_become_words | | FR-184, FR-185, NFR-013, TC-072 | 0.00 s |
| OK | test_the_four_documents_have_their_file_names | | FR-184, FR-185, NFR-013, TC-072 | 0.18 s |
| OK | test_the_full_documentation_has_a_contents_list_and_the_artefacts | | FR-184, FR-185, NFR-013, TC-072 | 0.39 s |
| OK | test_the_reader_finds_every_kind_of_block | | FR-184, FR-185, NFR-013, TC-072 | 0.04 s |
| OK | test_the_user_manual_is_built_from_the_real_document | | FR-184, FR-185, NFR-013, TC-072 | 7.55 s |

## tests/system/test_first_start.py — 3 tests, 3 passed

The first start on a new machine is one command that works from any folder

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_every_step_is_listed_from_any_folder_and_nothing_is_changed | | NFR-013, TC-071 | 1.42 s |
| OK | test_the_port_and_the_skipped_administrator_are_honoured | | NFR-013, TC-071 | 1.01 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_the_script_names_the_application_folder_it_works_in | | NFR-013, TC-071 | 1.02 s |

## tests/system/test_handover.py — 5 tests, 5 passed

The hand-over to the team repository (FR-188, scripts/handover.py): the audit, the clean export and

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_clean_tree_passes_the_audit | | FR-188, NFR-013, TC-075 | 0.46 s |
| OK | test_a_tag_that_does_not_exist_and_a_push_without_confirmation_are_refused | | FR-188, NFR-013, TC-075 | 0.87 s |
| OK | test_every_kind_of_finding_stops_the_handover | | FR-188, NFR-013, TC-075 | 0.94 s |
| OK | test_the_export_is_one_commit_by_the_named_author_with_the_tag | | FR-188, NFR-013, TC-075 | 1.03 s |
| OK | test_the_repository_carries_its_team_files | | FR-188, NFR-013, TC-075 | 0.04 s |

## tests/system/test_health.py — 1 tests, 1 passed

US-001 acceptance criterion: app boots, health endpoint returns 200.

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_health_returns_ok | | US-001 | 0.68 s |

## tests/system/test_i18n.py — 30 tests, 30 passed

I-22: the German interface. The catalogue must cover every string a template wraps in `()`,_

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_every_language_translates_the_whole_interface_inventory | | FR-155, FR-167, TC-049 | 0.76 s |
| OK | test_every_template_string_has_a_german_entry | | FR-155, FR-167, TC-049 | 0.93 s |
| OK | test_main_pages_render_in_each_language_without_leaking_format_markers[ar] | | FR-155, FR-167, TC-049 | 1.58 s |
| OK | test_main_pages_render_in_each_language_without_leaking_format_markers[bn] | | FR-155, FR-167, TC-049 | 1.42 s |
| OK | test_main_pages_render_in_each_language_without_leaking_format_markers[da] | | FR-155, FR-167, TC-049 | 1.53 s |
| OK | test_main_pages_render_in_each_language_without_leaking_format_markers[de] | | FR-155, FR-167, TC-049 | 1.57 s |
| OK | test_main_pages_render_in_each_language_without_leaking_format_markers[en] | | FR-155, FR-167, TC-049 | 1.53 s |
| OK | test_main_pages_render_in_each_language_without_leaking_format_markers[es] | | FR-155, FR-167, TC-049 | 1.56 s |
| OK | test_main_pages_render_in_each_language_without_leaking_format_markers[fr] | | FR-155, FR-167, TC-049 | 1.57 s |
| OK | test_main_pages_render_in_each_language_without_leaking_format_markers[gu] | | FR-155, FR-167, TC-049 | 2.59 s |
| OK | test_main_pages_render_in_each_language_without_leaking_format_markers[hi] | | FR-155, FR-167, TC-049 | 1.61 s |
| OK | test_main_pages_render_in_each_language_without_leaking_format_markers[it] | | FR-155, FR-167, TC-049 | 1.60 s |
| OK | test_main_pages_render_in_each_language_without_leaking_format_markers[ka] | | FR-155, FR-167, TC-049 | 1.60 s |
| OK | test_main_pages_render_in_each_language_without_leaking_format_markers[kn] | | FR-155, FR-167, TC-049 | 1.64 s |
| OK | test_main_pages_render_in_each_language_without_leaking_format_markers[ml] | | FR-155, FR-167, TC-049 | 1.61 s |
| OK | test_main_pages_render_in_each_language_without_leaking_format_markers[mr] | | FR-155, FR-167, TC-049 | 1.61 s |
| OK | test_main_pages_render_in_each_language_without_leaking_format_markers[or] | | FR-155, FR-167, TC-049 | 1.59 s |
| OK | test_main_pages_render_in_each_language_without_leaking_format_markers[pa] | | FR-155, FR-167, TC-049 | 2.58 s |
| OK | test_main_pages_render_in_each_language_without_leaking_format_markers[ta] | | FR-155, FR-167, TC-049 | 1.61 s |
| OK | test_main_pages_render_in_each_language_without_leaking_format_markers[te] | | FR-155, FR-167, TC-049 | 1.64 s |
| OK | test_main_pages_render_in_each_language_without_leaking_format_markers[ur] | | FR-155, FR-167, TC-049 | 1.61 s |
| OK | test_placeholders_survive_translation | | FR-155, FR-167, TC-049 | 0.60 s |
| OK | test_the_browser_language_picks_the_login_page_language | | FR-155, FR-167, TC-049 | 0.68 s |
| OK | test_the_catalogue_has_no_empty_or_identical_entries | | FR-155, FR-167, TC-049 | 0.57 s |
| OK | test_the_choice_before_login_applies_to_the_login_page | | FR-155, FR-167, TC-049 | 0.62 s |
| OK | test_the_german_login_page_keeps_english_credentials_hint_free_of_secrets | | FR-155, FR-167, TC-049 | 0.75 s |
| OK | test_the_language_choice_is_remembered_for_the_user | | FR-155, FR-167, TC-049 | 0.89 s |
| OK | test_the_user_preference_is_exposed_on_the_account_page | | FR-155, FR-167, TC-049 | 1.60 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_there_are_no_stale_german_entries_for_removed_template_strings | | FR-155, FR-167, TC-049 | 2.29 s |
| OK | test_translate_falls_back_to_the_english_text | | FR-155, FR-167, TC-049 | 0.62 s |

## tests/system/test_injection_and_session.py — 8 tests, 8 passed

TC-017, TC-026, TC-027, SEC-002, SEC-003, SEC-020-SEC-022: injection

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_audit_log_filters_resist_injection | | SEC-002, SEC-003, SEC-020, SEC-022, TC-017, TC-026, TC-027 | 0.81 s |
| OK | test_idle_session_expires_after_thirty_minutes | TC-017 / FR-002 / SEC-003. | FR-002, SEC-003, TC-017 | 0.82 s |
| OK | test_recent_activity_keeps_the_session_alive | | SEC-002, SEC-003, SEC-020, SEC-022, TC-017, TC-026, TC-027 | 1.83 s |
| OK | test_server_side_validation_ignores_client_side_constraints | SEC-020: every rule is enforced server-side, whatever the browser did. | SEC-020 | 1.16 s |
| OK | test_session_cookie_attributes | SEC-002 (the Secure flag is off only in the dev/test configs). | SEC-002 | 0.78 s |
| OK | test_session_expires_at_the_absolute_lifetime | SEC-003: 12 hours regardless of activity. | SEC-003 | 0.77 s |
| OK | test_sql_injection_probes_have_no_effect | TC-027 / SEC-021. | SEC-021, TC-027 | 1.51 s |
| OK | test_user_supplied_text_is_escaped_everywhere_it_renders | TC-026 / SEC-022. | SEC-022, TC-026 | 2.30 s |

## tests/system/test_logging_and_passwords.py — 8 tests, 8 passed

Security policy details from 11-security-architecture.md: §4 password policy

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_administrator_reset_also_honours_history | | | 1.20 s |
| OK | test_authenticated_requests_record_the_user_id | | | 0.75 s |
| OK | test_blocklist_ignores_comments_and_short_entries | | | 0.00 s |
| OK | test_last_five_passwords_cannot_be_reused | | | 3.97 s |
| OK | test_only_hashes_are_stored_in_the_history | | | 0.84 s |
| OK | test_policy_length_and_blocklist | | | 0.00 s |
| OK | test_requests_are_logged_as_json_without_credentials | | | 0.89 s |
| OK | test_unhandled_errors_are_logged_with_the_request_id | | | 0.80 s |

## tests/system/test_multiprocess.py — 4 tests, 4 passed

Two real server processes on one database file (docs/14-deployment-architecture.md §3):

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_concurrent_writers_in_two_processes_all_succeed | | NFR-009, TC-050 | 29.17 s |
| OK | test_the_ip_limit_is_shared_and_other_users_are_unaffected_until_it_is_reached | | NFR-009, TC-050 | 26.67 s |
| OK | test_the_limit_survives_a_restart | | NFR-009, TC-050 | 27.35 s |
| OK | test_the_username_limit_is_shared_by_both_processes | | NFR-009, TC-050 | 26.76 s |

## tests/system/test_no_account_names.py — 2 tests, 2 passed

The documents and the guide do not name the Git account or the repository (a decision of the

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_no_document_or_script_names_the_account_or_the_repository | | NFR-013 | 1.01 s |
| OK | test_the_guide_says_who_to_ask_for_the_repository | | NFR-013 | 0.01 s |

## tests/system/test_no_cloud.py — 3 tests, 3 passed

US-120 / TC-029: block cloud-SDK imports, cloud identity providers and

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_no_cloud_sdk_imports_in_source | | TC-029, US-120 | 1.53 s |
| OK | test_no_cloud_sdk_in_requirements | | TC-029, US-120 | 0.01 s |
| OK | test_templates_and_static_reference_no_external_assets | | TC-029, US-120 | 0.51 s |

## tests/system/test_openapi_conformance.py — 5 tests, 5 passed

EPIC-11 / FR-080: the code and docs/artifacts/openapi.yaml describe the

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_every_documented_operation_is_implemented_and_vice_versa | | FR-080 | 1.86 s |
| OK | test_live_payloads_only_use_declared_properties | | FR-080 | 1.51 s |
| OK | test_lookup_and_domain_request_payloads_match_contract | | FR-080 | 1.31 s |
| OK | test_revision_diff_payload_matches_contract | | FR-080 | 1.36 s |
| OK | test_unauthenticated_requests_are_rejected_for_every_documented_operation | | FR-080 | 0.77 s |

## tests/system/test_permission_grid.py — 14 tests, 14 passed

TC-018 / TC-018a / TC-018b / TC-019 / FR-081 / SEC-010: the full

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_api_endpoints_require_authentication | | FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019 | 1.29 s |
| OK | test_api_permission_grid[ADMINISTRATOR] | | FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019 | 1.95 s |
| OK | test_api_permission_grid[CHIEF_ENGINEER] | | FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019 | 1.81 s |
| OK | test_api_permission_grid[ENGINEER] | | FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019 | 1.74 s |
| OK | test_api_permission_grid[REVIEWER] | | FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019 | 2.77 s |
| OK | test_api_permission_grid[STCD_OWNER] | | FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019 | 1.70 s |
| OK | test_api_permission_grid[VIEWER] | | FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019 | 1.70 s |
| OK | test_ui_permission_grid[ADMINISTRATOR] | | FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019 | 3.02 s |
| OK | test_ui_permission_grid[CHIEF_ENGINEER] | | FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019 | 2.05 s |
| OK | test_ui_permission_grid[ENGINEER] | | FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019 | 3.02 s |
| OK | test_ui_permission_grid[REVIEWER] | | FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019 | 2.01 s |
| OK | test_ui_permission_grid[STCD_OWNER] | | FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019 | 1.66 s |
| OK | test_ui_permission_grid[VIEWER] | | FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019 | 1.40 s |
| OK | test_ui_redirects_anonymous_users_to_login | | FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019 | 1.29 s |

## tests/system/test_production_config.py — 7 tests, 7 passed

SEC-002 / deployment hardening: a production instance must not start with a

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_development_and_testing_still_start_with_the_default_key | | SEC-002 | 0.34 s |
| OK | test_production_refuses_a_weak_secret_key[] | | SEC-002 | 0.01 s |
| OK | test_production_refuses_a_weak_secret_key[change-me-in-every-real-environment] | | SEC-002 | 0.00 s |
| OK | test_production_refuses_a_weak_secret_key[dev-secret-key-change-me] | | SEC-002 | 0.01 s |
| OK | test_production_refuses_a_weak_secret_key[too-short] | | SEC-002 | 0.00 s |
| OK | test_production_refuses_a_weak_secret_key[xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx] | | SEC-002 | 0.00 s |
| OK | test_production_starts_with_a_strong_secret_key | | SEC-002 | 0.17 s |

## tests/system/test_route_inventory.py — 13 tests, 13 passed

Every route is access-controlled, and the declared role lists are what is enforced.

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_tokenised_endpoint_is_always_a_get | API tokens may only ever read (docs/11 §12): allow_token must never sit on a write. | FR-081, SEC-010, TC-018, TC-048 | 0.63 s |
| OK | test_every_gated_route_refuses_anonymous_callers | | FR-081, SEC-010, TC-018, TC-048 | 0.86 s |
| OK | test_every_route_declares_its_access_control | | FR-081, SEC-010, TC-018, TC-048 | 0.60 s |
| OK | test_login_only_routes_are_reachable_by_every_role | | FR-081, SEC-010, TC-018, TC-048 | 6.80 s |
| OK | test_login_page_credentials_still_work_for_the_ui_walk | | FR-081, SEC-010, TC-018, TC-048 | 0.76 s |
| OK | test_role_lists_are_never_empty_and_only_name_real_roles | | FR-081, SEC-010, TC-018, TC-048 | 1.67 s |
| OK | test_the_declared_role_list_is_what_each_role_gets[ADMINISTRATOR] | | FR-081, SEC-010, TC-018, TC-048 | 6.88 s |
| OK | test_the_declared_role_list_is_what_each_role_gets[CHIEF_ENGINEER] | | FR-081, SEC-010, TC-018, TC-048 | 3.01 s |
| OK | test_the_declared_role_list_is_what_each_role_gets[ENGINEER] | | FR-081, SEC-010, TC-018, TC-048 | 3.23 s |
| OK | test_the_declared_role_list_is_what_each_role_gets[REVIEWER] | | FR-081, SEC-010, TC-018, TC-048 | 3.08 s |
| OK | test_the_declared_role_list_is_what_each_role_gets[STCD_OWNER] | | FR-081, SEC-010, TC-018, TC-048 | 3.85 s |
| OK | test_the_declared_role_list_is_what_each_role_gets[VIEWER] | | FR-081, SEC-010, TC-018, TC-048 | 2.37 s |
| OK | test_the_public_list_is_exactly_the_routes_that_need_no_session | | FR-081, SEC-010, TC-018, TC-048 | 0.60 s |

## tests/system/test_security_headers.py — 4 tests, 4 passed

SEC-025: standard security headers on every response.

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_api_response_has_security_headers | | SEC-025 | 0.58 s |
| OK | test_hsts_sent_over_https | | SEC-025 | 0.60 s |
| OK | test_templates_have_no_inline_script_or_handlers | The CSP forbids inline script; guard against regressions. | SEC-025 | 0.31 s |
| OK | test_ui_page_has_csp_and_no_hsts_over_plain_http | | SEC-025 | 0.72 s |

## tests/system/test_showcase.py — 6 tests, 6 passed

_The design showcase decks (FR-186, scripts/build_showcase.py): the slide model and its layout_

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_text_that_does_not_fit_stops_the_build | | FR-186, FR-187, TC-073 | 0.16 s |
| OK | test_both_decks_are_built_from_the_screenshots_and_every_picture_is_described | | FR-186, FR-187, TC-073 | 0.42 s |
| OK | test_the_browser_is_found_or_named | | FR-186, FR-187, TC-073 | 0.01 s |
| OK | test_the_decks_are_written_as_powerpoint_files | | FR-186, FR-187, TC-073 | 4.30 s |
| OK | test_the_demo_has_its_data_and_the_application_has_no_demo_routes | | FR-186, FR-187, TC-073 | 8.22 s |
| OK | test_the_script_refuses_an_unknown_deck | | FR-186, FR-187, TC-073 | 0.03 s |

## tests/system/test_sqlite_path.py — 2 tests, 2 passed

A new machine: the database file is found, and its folder created, from the example .env

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_relative_sqlite_path_is_read_from_the_application_directory | | NFR-013 | 0.22 s |
| OK | test_an_absolute_path_gets_its_folder_and_memory_is_left_alone | | NFR-013 | 0.36 s |

## tests/system/test_translations_doc.py — 27 tests, 27 passed

The translation catalogue document and the review sheets for native speakers

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_every_language_has_its_section_and_its_coverage_row | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 0.11 s |
| OK | test_right_to_left_languages_get_a_right_to_left_translation_column[ar-right] | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 0.38 s |
| OK | test_right_to_left_languages_get_a_right_to_left_translation_column[fr-left] | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 0.42 s |
| OK | test_right_to_left_languages_get_a_right_to_left_translation_column[ur-right] | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 1.24 s |
| OK | test_string_ids_are_stable_and_never_collide | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 0.01 s |
| OK | test_the_command_line_writes_one_sheet_per_language_and_refuses_unknown_ones | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 5.58 s |
| OK | test_the_concept_names_the_layers_the_checks_and_the_open_items | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 0.02 s |
| OK | test_the_document_is_exactly_what_the_generator_produces_from_the_code | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 0.47 s |
| OK | test_the_fingerprint_changes_when_one_translation_changes | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 0.01 s |
| OK | test_the_review_sheet_of_every_language_is_complete[ar] | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 0.48 s |
| OK | test_the_review_sheet_of_every_language_is_complete[bn] | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 0.45 s |
| OK | test_the_review_sheet_of_every_language_is_complete[da] | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 0.47 s |
| OK | test_the_review_sheet_of_every_language_is_complete[de] | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 0.45 s |
| OK | test_the_review_sheet_of_every_language_is_complete[es] | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 1.28 s |
| OK | test_the_review_sheet_of_every_language_is_complete[fr] | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 0.44 s |
| OK | test_the_review_sheet_of_every_language_is_complete[gu] | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 1.41 s |
| OK | test_the_review_sheet_of_every_language_is_complete[hi] | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 1.35 s |
| OK | test_the_review_sheet_of_every_language_is_complete[it] | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 0.45 s |
| OK | test_the_review_sheet_of_every_language_is_complete[ka] | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 0.59 s |
| OK | test_the_review_sheet_of_every_language_is_complete[kn] | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 0.49 s |
| OK | test_the_review_sheet_of_every_language_is_complete[ml] | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 0.46 s |
| OK | test_the_review_sheet_of_every_language_is_complete[mr] | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 0.47 s |
| OK | test_the_review_sheet_of_every_language_is_complete[or] | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 0.45 s |
| OK | test_the_review_sheet_of_every_language_is_complete[pa] | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 0.42 s |
| OK | test_the_review_sheet_of_every_language_is_complete[ta] | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 0.44 s |
| OK | test_the_review_sheet_of_every_language_is_complete[te] | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 0.44 s |
| OK | test_the_review_sheet_of_every_language_is_complete[ur] | | FR-110, FR-111, FR-155, FR-157, FR-161, TC-057 | 0.55 s |

## tests/system/test_uat.py — 14 tests, 14 passed

TC-030: UAT-01 .. UAT-13 (docs/13-test-strategy.md §7) as scripted

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_uat_01_new_code_has_no_identifier_until_submitted | | UAT-01 | 2.64 s |
| OK | test_uat_02_either_signoff_order_reaches_pending_approval_automatically | | UAT-02 | 1.71 s |
| OK | test_uat_03_chief_engineer_approval_visible_to_the_engineer | | UAT-03 | 2.11 s |
| OK | test_uat_04_rejection_comment_visible_and_prior_signoff_cleared | | UAT-04 | 2.11 s |
| OK | test_uat_05_cross_domain_signoff_unblocks_submit | | UAT-05 | 1.50 s |
| OK | test_uat_06_release_build_excludes_unselected_and_sandbox | | UAT-06 | 2.90 s |
| OK | test_uat_07_viewer_searches_and_exports_released_catalogue_for_one_group | | UAT-07 | 2.26 s |
| OK | test_uat_08_sandbox_lifecycle_and_traceable_deletion | | UAT-08 | 1.26 s |
| OK | test_uat_09_nobody_can_decide_their_own_submission | | UAT-09 | 2.62 s |
| OK | test_uat_09b_chief_engineer_cannot_be_a_prior_signer_or_author | | TC-030, UAT-01, UAT-13 | 2.08 s |
| OK | test_uat_10_full_database_export_is_administrator_only | | UAT-10 | 1.61 s |
| OK | test_uat_11_import_creates_only_governed_drafts_and_an_error_report | | UAT-11 | 1.29 s |
| OK | test_uat_12_subgroup_is_mandatory_and_overlaps_are_refused | | UAT-12 | 1.22 s |
| OK | test_uat_13_failed_form_keeps_input_and_dropdowns_explain_themselves | | UAT-13 | 1.50 s |

## tests/system/test_version.py — 4 tests, 4 passed

The version of the application is written once and shown where a person or a monitor looks for

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_the_api_contract_documents_the_version_in_both_health_answers | | FR-184, NFR-013, TC-049 | 0.03 s |
| OK | test_the_health_endpoints_name_the_version | | FR-184, NFR-013, TC-049 | 0.82 s |
| OK | test_the_login_page_and_the_menu_show_the_version | | FR-184, NFR-013, TC-049 | 1.58 s |
| OK | test_the_version_is_a_release_number_and_pyproject_agrees | | FR-184, NFR-013, TC-049 | 0.04 s |

## tests/system/test_windows_gateway.py — 7 tests, 7 passed

The Windows sign-in against a real Windows identity (docs/03-srs.md FR-163, docs/11 SEC-033).

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_a_header_the_client_sends_is_dropped_by_the_gateway | | FR-163, SEC-033, TC-062 | 1.19 s |
| OK | test_a_request_without_credentials_gets_a_challenge_and_never_reaches_the_application | | FR-163, SEC-033, TC-062 | 1.70 s |
| OK | test_an_unknown_windows_user_and_a_deactivated_account_are_refused | | FR-163, SEC-033, TC-062 | 2.48 s |
| OK | test_sspi_names_the_windows_user_who_runs_the_process[NTLM] | | FR-163, SEC-033, TC-062 | 0.01 s |
| OK | test_sspi_names_the_windows_user_who_runs_the_process[Negotiate] | | FR-163, SEC-033, TC-062 | 0.00 s |
| OK | test_the_header_is_ignored_when_it_does_not_come_from_the_trusted_gateway | | FR-163, SEC-033, TC-062 | 1.58 s |
| OK | test_the_whole_chain_signs_the_windows_user_in | | FR-163, SEC-033, TC-062 | 1.24 s |

## tests/performance/test_performance.py — 9 tests, 9 passed

NFR-001, TC-024, TC-025, NFR-004: performance targets on a temp-file SQLite

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_concurrent_identifier_allocation_in_one_group | TC-025: parallel submissions in one popular group yield unique, gap-free | TC-025 | 1.14 s |
| OK | test_export_issues_a_constant_number_of_queries | Regression guard for the N+1 problem: exporting 500 revisions must not cost | NFR-001, NFR-004, TC-024, TC-025 | 1.18 s |
| OK | test_export_of_5000_records_within_target[csv-5] | NFR-001: CSV/JSON <= 5 s, PDF <= 15 s for 5,000 records (XLSX held to the PDF limit). | NFR-001 | 2.50 s |

| Result | Test | Purpose | Requirements | Time |
|---|---|---|---|---|
| OK | test_export_of_5000_records_within_target[json-5] | NFR-001: CSV/JSON <= 5 s, PDF <= 15 s for 5,000 records (XLSX held to the PDF limit). | NFR-001 | 2.75 s |
| OK | test_export_of_5000_records_within_target[pdf-15] | NFR-001: CSV/JSON <= 5 s, PDF <= 15 s for 5,000 records (XLSX held to the PDF limit). | NFR-001 | 10.63 s |
| OK | test_export_of_5000_records_within_target[xlsx-15] | NFR-001: CSV/JSON <= 5 s, PDF <= 15 s for 5,000 records (XLSX held to the PDF limit). | NFR-001 | 4.68 s |
| OK | test_fifty_concurrent_users_are_served_without_errors | NFR-004: 50 concurrent named users (each searching every ~8 s — | NFR-004 | 30.59 s |
| OK | test_reads_are_not_blocked_by_an_open_write_transaction | WAL read concurrency: a reader proceeds while another connection holds | NFR-001, NFR-004, TC-024, TC-025 | 1.30 s |
| OK | test_search_and_read_latency_at_20000_revisions | TC-024. | TC-024 | 12.69 s |

# 9. Run history

| # | Date (UTC) | Commit | Order | Tests | Passed | Failed | Retried | Coverage | Gates | Duration |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2026-10-21 16:36 | f23ef18* | reverse | 318 | 318 | 0 | 1 | 88% | 5/5 pass | 232 s |
| 2 | 2026-10-21 16:42 | f23ef18* | shuffle:11 | 318 | 318 | 0 | 0 | 88% | 5/5 pass | 235 s |
| 3 | 2026-10-21 16:49 | f23ef18* | file | 318 | 318 | 0 | 1 | 88% | 5/5 pass | 237 s |
| 4 | 2026-10-21 16:57 | f23ef18* | file | 318 | 318 | 0 | 0 | 88% | 5/5 pass | 244 s |
| 5 | 2026-10-21 17:09 | f23ef18* | shuffle:23 | 319 | 319 | 0 | 0 | 88% | 5/5 pass | 242 s |
| 6 | 2026-10-21 17:15 | f23ef18* | file | 319 | 319 | 0 | 0 | 88% | 5/5 pass | 237 s |
| 7 | 2026-10-22 10:02 | f23ef18* | file | 329 | 329 | 0 | 0 | 89% | 5/5 pass | 150 s |
| 8 | 2026-10-23 06:16 | c0aff94* | file | 601 | 600 | 1 | 0 | 92% | 5/5 pass | 589 s |
| 9 | 2026-10-23 06:27 | c0aff94* | reverse | 601 | 600 | 1 | 0 | 92% | 5/5 pass | 526 s |
| 10 | 2026-10-23 06:38 | c0aff94* | shuffle:7 | 601 | 600 | 1 | 0 | 92% | 5/5 pass | 473 s |
| 11 | 2026-10-23 06:50 | c0aff94* | file | 602 | 602 | 0 | 0 | 92% | 5/5 pass | 472 s |
| 12 | 2026-10-23 06:59 | c0aff94* | reverse | 602 | 602 | 0 | 0 | 92% | 5/5 pass | 470 s |
| 13 | 2026-10-23 07:08 | c0aff94* | shuffle:7 | 602 | 602 | 0 | 0 | 92% | 5/5 pass | 469 s |
| 14 | 2026-10-23 07:18 | c0aff94* | file+filedb | 602 | 601 | 1 | 0 | 92% | 5/5 pass | 542 s |
| 15 | 2026-10-23 07:31 | c0aff94* | file+filedb | 602 | 602 | 0 | 0 | 92% | 5/5 pass | 590 s |
| 16 | 2026-10-23 07:41 | c0aff94* | file | 602 | 602 | 0 | 0 | 92% | 5/5 pass | 505 s |
| 17 | 2026-10-23 07:54 | c0aff94* | reverse | 607 | 607 | 0 | 0 | 92% | 5/5 pass | 459 s |
| 18 | 2026-10-23 08:04 | c0aff94* | shuffle:7 | 607 | 607 | 0 | 0 | 92% | 5/5 pass | 491 s |
| 19 | 2026-10-23 08:16 | c0aff94* | file+filedb | 607 | 607 | 0 | 0 | 92% | 5/5 pass | 628 s |
| 20 | 2026-10-23 08:25 | c0aff94* | file | 607 | 607 | 0 | 0 | 92% | 5/5 pass | 508 s |
| 21 | 2026-10-23 15:33 | c0aff94* | file | 807 | 807 | 0 | 0 | 92% | 5/5 pass | 685 s |
| 22 | 2026-10-23 19:53 | 3670346* | file | 830 | 828 | 2 | 0 | 93% | 2/5 pass | 850 s |
| 23 | 2026-10-23 20:10 | 3670346* | file | 830 | 830 | 0 | 0 | 93% | 5/5 pass | 728 s |
| 24 | 2026-10-23 23:30 | 9f038bc* | file | 848 | 847 | 1 | 0 | 93% | 5/5 pass | 741 s |
| 25 | 2026-10-23 23:47 | 9f038bc* | file | 848 | 848 | 0 | 0 | 93% | 5/5 pass | 748 s |
| 26 | 2026-10-24 10:42 | e07a82d* | file | 867 | 867 | 0 | 0 | 93% | 4/5 pass | 768 s |
| 27 | 2026-10-24 10:57 | 88bf318* | file | 867 | 867 | 0 | 0 | 93% | 5/5 pass | 772 s |
| 28 | 2026-10-24 11:50 | 9828815* | file | 876 | 876 | 0 | 0 | 93% | 5/5 pass | 773 s |
| 29 | 2026-10-24 12:42 | 1f1f0c9* | file | 892 | 891 | 1 | 0 | 93% | 5/5 pass | 807 s |
| 30 | 2026-10-24 12:58 | 1f1f0c9* | file | 892 | 892 | 0 | 0 | 93% | 5/5 pass | 806 s |
| 31 | 2026-10-24 14:37 | a81555a* | file | 904 | 904 | 0 | 0 | 93% | 4/5 pass | 967 s |
| 32 | 2026-10-24 14:56 | a81555a* | file | 915 | 915 | 0 | 0 | 93% | 5/5 pass | 979 s |
| 33 | 2026-10-24 16:00 | 2cd8fe6* | file | 927 | 927 | 0 | 0 | 93% | 5/5 pass | 993 s |
| 34 | 2026-10-24 17:31 | afc84a3* | file | 946 | 945 | 1 | 0 | 93% | 3/5 pass | 1051 s |
| 35 | 2026-10-24 17:53 | afc84a3* | file | 946 | 943 | 3 | 0 | 93% | 5/5 pass | 1050 s |
| 36 | 2026-10-24 18:12 | afc84a3* | file | 946 | 946 | 0 | 0 | 93% | 5/5 pass | 1054 s |
| 37 | 2026-10-25 11:41 | 016e74f* | file | 956 | 956 | 0 | 0 | 93% | 3/5 pass | 1207 s |
| 38 | 2026-10-25 12:06 | 016e74f* | file | 956 | 956 | 0 | 0 | 93% | 5/5 pass | 1205 s |

| # | Date (UTC) | Commit | Order | Tests | Passed | Failed | Retried | Coverage | Gates | Duration |
|---|---|---|---|---|---|---|---|---|---|---|
| 39 | 2026-10-26 09:59 | b0029bd* | file | 958 | 958 | 0 | 0 | 93% | 5/5 pass | 1163 s |
| 40 | 2026-10-26 10:26 | 2a5a8ff | file | 966 | 966 | 0 | 0 | 93% | 5/5 pass | 1113 s |
| 41 | 2026-10-26 11:39 | 97af72c | file | 973 | 973 | 0 | 0 | 93% | 5/5 pass | 1099 s |
| 42 | 2026-10-26 12:29 | a62d0c6* | file | 1007 | 1007 | 0 | 0 | 93% | 5/5 pass | 1209 s |
| 43 | 2026-10-26 13:11 | 613aa61* | file | 1020 | 1019 | 0 | 0 | 93% | 5/5 pass | 1187 s |
| 44 | 2026-10-26 13:44 | 3b18219 | file | 1025 | 1023 | 1 | 0 | 93% | 5/5 pass | 1194 s |
| 45 | 2026-10-26 14:07 | 5094e00 | file | 1025 | 1025 | 0 | 0 | 93% | 5/5 pass | 1205 s |
| 46 | 2026-10-27 16:23 | 45bd50e* | file | 1035 | 1030 | 5 | 0 | 93% | 4/5 pass | 1292 s |
| 47 | 2026-10-27 16:51 | 45bd50e* | file | 1035 | 1035 | 0 | 0 | 93% | 4/5 pass | 1285 s |
| 48 | 2026-10-27 17:16 | 45bd50e* | file | 1035 | 1035 | 0 | 0 | 93% | 5/5 pass | 1292 s |
| 49 | 2026-10-27 19:40 | 78dcd55* | file | 1079 | 1077 | 2 | 0 | 93% | 5/5 pass | 1409 s |
| 50 | 2026-10-27 20:09 | 78dcd55* | file | 1079 | 1079 | 0 | 0 | 93% | 5/5 pass | 1421 s |
| 51 | 2026-10-28 19:13 | a696bc6* | file | 1082 | 1080 | 2 | 0 | 93% | 5/5 pass | 2077 s |
| 52 | 2026-10-28 19:51 | a696bc6* | file | 1082 | 1082 | 0 | 0 | 93% | 5/5 pass | 1983 s |
| 53 | 2026-10-28 20:57 | 0a54cda* | file | 1089 | 1089 | 0 | 0 | 93% | 5/5 pass | 2203 s |
| 54 | 2026-10-28 23:41 | a741076* | file | 1282 | 1279 | 0 | 0 | 93% | 5/5 pass | 2155 s |
| 55 | 2026-10-29 00:25 | a741076* | shuffle:11 | 1282 | 1279 | 0 | 0 | 93% | 5/5 pass | 2139 s |
| 56 | 2026-10-29 01:18 | a741076* | file | 1283 | 1280 | 0 | 0 | 93% | 5/5 pass | 1749 s |